

Rock and a Hard Place: Attack Hardness in Neural Network-assisted Side Channel Analysis

Syedmohammad Nouraniboosjin and Fatameh Ganji

Worcester Polytechnic Institute, Worcester, USA
snouraniboosjin@wpi.edu, fganji@wpi.edu

Abstract. Side-channel analysis (SCA) has become a crucial area in ensuring the security of hardware implementations against potential vulnerabilities. With advancements in machine learning (ML) and artificial intelligence (AI), neural network(NN)-assisted techniques for SCA have demonstrated significant effectiveness. However, a fundamental question remains unanswered: are traces corresponding to different subkeys equally hard to attack? This paper addresses this issue by leveraging explainable AI techniques to analyze the hardness levels and, consequently, the root cause of hardness. To systematically investigate this problem, we reintroduce hardness metrics in SCA, which have been known to the ML community. Those metrics include query hardness (QH), log odds (LO), and matrix-based Rényi entropy (MRE). The challenge in this regard is that (virtually all) hardness metrics in ML cannot be adopted as they are. This is because ML and SCA metrics have conflicting goals, namely, boosting accuracy and rank. Through careful experimentation, we identify the shortcomings of QH and LO in SCA and recommend MRE as a suitable hardness metric for SCA.

We also study how hardness has been seen in SCA, where recent work has suggested the influence of class “labels” on the attack difficulty. Employing rigorous evaluation, our paper demonstrates that no statistically significant evidence can be found to support this claim. This leads us to the question of how much traces’ time samples affect the inherent hardness in distinguishing key candidates. Our novel explainable AI (XAI) approach not only answers this, but also makes a link between hardness and rank as the common performance metric in SCA. Our findings indicate that hardness values are influenced mainly by time samples rather than specific key labels. Furthermore, we examine whether hardness captures intrinsic properties of the traces, specifically, their lack of separability in feature space due to their inherent similarities. To this end, we introduce, for the first time in the context of SCA, the use of maximum mean discrepancy (MMD) as a principled metric. MMD effectively links trace hardness with attack difficulty by quantifying distributional differences induced by traces’ time samples. In addition to visualization techniques showing the root cause of hardness based on MMD, we employ XAI to explain the connection between MMD and attack hardness. Our proposed methodology enhances the understanding of attack difficulty in SCA and contributes to developing more robust evaluation metrics for profiling attacks.

Keywords: Deep learning · Neural networks · Side-channel analysis · Metrics · Explainable AI (XAI) · Hardness · Matrix-based Rényi Entropy · Maximum Mean Discrepancy

1 Introduction

Side-channel analysis (SCA) is widely acknowledged as one of the most effective attack methods against cryptographic implementations. The core principle of SCA is that an

adversary can infer sensitive information by observing and measuring physical effects (e.g., power consumption and electromagnetic emissions) during program execution [KJJ99, GMO01]. The worst-case scenario in the context of SCA refers to an adversary capable of measuring side channels from the target device with complete control over all inputs (such as plaintexts or ciphertexts) as well as over secret parameters, including keys. This scenario represents an adversary who can practice (virtually unlimited) *profiling* in order to conduct the most powerful attack possible, cf. [ABB⁺20]. The goal of the profiling phase is to characterize the statistical relationship between the sensitive variables and the observed side-channel leakage. The profiling step is followed by the attack phase, where the obtained knowledge about this relationship is used to distinguish the secret. Template attacks [CRR02] were the first examples of such profiled analysis. Relying on modeling the target intermediate variables as Gaussian mixtures [PR09, PR10, SVCO⁺10, VCMKS12], Gaussian templates can be created to launch the attack. Template attacks are considered information-theoretically optimal, given that a sufficiently large number of traces are available and the underlying Gaussian assumption *holds* [MDP20]. In the absence of a sufficient number of traces or the presence of countermeasures, estimating the probability distribution of observed side-channel leakage conditioned on the sensitive variables can be handled by alternative solutions. A prime example of such solutions is neural networks (NNs) [MZ13, LBM14, MDM16].

To learn the conditional distribution, NNs implicitly employ structured relationships between inputs. Metric learning aims to quantify such a relationship by selecting/learning an optimal distance metric tailored to the specific learning task [KB19]. Distance metrics that lack strong generalization to the problem are unlikely to yield effective results. Consequently, selecting an appropriate distance metric is essential for achieving reliable performance on the input data. One of the fundamental similarity metrics commonly used for classification is Euclidean distance, widely adopted in SCA-related literature [GLRP06, SMY09, DPRS11, MDP20, BCGR24].

Even after selecting the proper distance metric, the classification task can still be *hard*. Side-channel practitioners, including industrial developers and evaluators as well as academic researchers, mainly relate hardness to what is known or unknown by the adversary during the profiling phase. On the one hand, the worst-case threat model assumes that the adversary knows the values of the random nonces in countermeasures, e.g., masking or shuffling [ABB⁺20]. On the other hand, an uninformed threat model restricts this knowledge from leveraging automation and being closer to a real-world adversary, cf. [MCLS23]. Besides this, a line of research has been devoted to tuning hyperparameters as a challenging task in NN-assisted SCA [PPM⁺21, WPP22, LW22, RWPP21, AGF23]. [MCLS23] has also argued that the main difficulty for the adversary under non-worst case scenario is due to the drawbacks of gradient decent-based optimization procedures. For this purpose, that paper relied on a theorem [SSSS17], which assumes that the leakage distribution could be noise-free and the same for each measurement instance, which may not hold in practice.

From another perspective, a research avenue has considered the influence of labels on the hardness of classification [WPP21, WWK⁺23]. For instance, [WPP21] suggests that a smaller label distance may indicate similar leakage traces, hence making the class prediction (key extraction) harder for the NN. In another attempt, [WWK⁺23] has assumed that a label closer to the correct label is more likely to be selected. Under this assumption, that paper has proposed label smoothing to transform one-hot encoded labels to Gaussian-distributed labels. Label smoothing implicitly calibrates learned models by adjusting their output confidences to better align with the actual accuracies of their predictions [MKH19]. It encourages training examples from the same class to form tightly clustered representations, which leads to a loss of information in the logits regarding similarities between instances of different classes, cf. [MKH19]. Hence, the attack performance

could not be improved compared to other model calibration techniques, e.g., temperature calibration [NG24]. More importantly, the attack can become even harder when losing the information regarding similarities between traces with different labels.

Contribution. Motivated by the lack of thorough analysis of attack hardness, this paper aims to close the gap between the foundation of NN computation and SCA. In fact, the notion of hardness has already been formulated in the former, whereas its applicability in the latter field has remained unexplored. Needless to say, not all hardness metrics devised in ML literature could be useful in SCA due to the difference in the ultimate goal of ML and SCA. While the objective of ML approaches is to enhance NNs’ performance in terms of accuracy, SCA incentivizes highly ranked predictions made by NNs. Therefore, our paper focused on bringing these two fields together by contributing to the following.

- We systematically explore hardness metrics suggested in SCA and NN computation in order to determine shortcomings of ones in SCA and advantages offered by the hardness metrics as defined in NN computation-related literature. A key challenge in this context is that nearly all existing hardness metrics from ML are not directly applicable to SCA. This arises from the differing objectives of the two domains: ML aims to maximize classification accuracy, whereas SCA focuses on minimizing the rank of the correct key. Through systematic experimentation, we highlight the limitations of mainstream hardness metrics within the SCA setting and propose matrix-based Rényi entropy as an appropriate measure of hardness that aligns with the goals of SCA.
- We further investigate the concept of hardness as it has been interpreted in SCA, where recent studies have proposed that class “labels” may influence attack difficulty. Through a comprehensive evaluation, we demonstrate that no statistically significant evidence supports this hypothesis. This finding motivates a deeper analysis of how the time samples in leakage traces contribute to the inherent difficulty in distinguishing between key candidates. To this end, we introduce a novel explainable AI (XAI)-based framework that not only provides insights into this relationship but also establishes a connection between hardness and key rank, i.e., one of the most widely used performance metrics in SCA.
- Finally, we investigate whether hardness reflects intrinsic characteristics of traces, namely, their limited separability in feature space due to inherent similarities. To address this, we propose, for the first time in the context of side-channel analysis (SCA), the application of maximum mean discrepancy (MMD) as a rigorous statistical measure. MMD enables a quantitative connection between trace hardness and attack difficulty due to the traces’ time sample. Beyond t-SNE [VdMH08] and UMAP [MHM18] visual analysis to identify the sources of hardness via MMD, we also leverage XAI techniques to explain the relationship between MMD and the observed attack hardness.

2 Background

2.1 Notations

We use calligraphic letters, such as $\mathcal{X}$, to represent sets. Random variables are denoted by uppercase letters like X , while random vectors are represented by bold uppercase letters such as $\mathbf{X}$. The realizations of random variables are indicated by lowercase letters like x , with vectors represented by bold lowercase letters $\mathbf{x}$. Matrices are denoted using bold capital letters. For a vector $\mathbf{x}$, its t -th component is expressed as $\mathbf{x}[t]$, while x_i denotes the i -th observation of a random variable X . The probability density function (pdf) of a

continuous random variable X is denoted by f_X , where $f_X(x) = p(X = x)$. The expected value is represented by $\mathbb{E}[\cdot]$, with possible subscripts for clarification, such as $\mathbb{E}_X[\cdot]$.

Shannon conditional entropy is formulated as $H(y | x) = H(y, x) - H(x)$. Mutual Information (MI) and Conditional Mutual Information (CMI) are defined as $I(x; y) = H(x) + H(y) - H(x, y)$, $I(x; y | x') = H(x, x') + H(y, x') - H(x, y, x') - H(x')$. Here, H denotes entropy or joint entropy $H = - \sum_{x \in X} p(x) \log p(x)$, where $p(x)$ is the probability of the random variable X taking the value x .

2.2 Profiled Side-Channel Analysis

Side-Channel Analysis (SCA) uses observations $\mathbf{L}$ made during algorithm execution on hardware to extract secret information [BPS⁺18]. When an adversary gains access to an open device, where manipulation of inputs and secrets is possible, they can conduct profiling attacks.

A profiling SCA attack comprises two phases: a profiling phase and an attack phase [BPS⁺18].

For each candidate key $k \in \mathcal{K}$, the attacker computes an estimated probability distribution $\hat{g}_k$:

$$g_k : (\ell, p) \mapsto p(\mathbf{L} = \ell \mid (P, K) = (p, k)). \quad (1)$$

The estimation $\hat{g}_k$ is derived from a profiling set $\mathcal{D}_{\text{profiling}} \doteq \{(\ell_i, p_i, k_i)\}_{i=1}^{N_p}$, where N_p is the number of traces collected under known plaintexts and keys. Trace set and input set are denoted by $\mathcal{L}$ and $\mathcal{X}$, respectively.

In the attack phase, the attacker gathers $\mathcal{D}_{\text{attack}} \doteq \{(\ell_i, p_i)\}_{i=1}^{N_a}$, where k^* is the undisclosed fixed key. The attacker computes a maximum-likelihood score for each key:

$$\mathbf{d}_{Na}[k] = \prod_{i=1}^{N_a} \Pr((P, K) = (p_i, k) \mid \mathbf{L} = \ell_i) = \prod_{i=1}^{N_a} \frac{\Pr(\mathbf{L} = \ell_i \mid (P, K) = (p_i, k)) f_{\mathbf{L}}(\ell_i)}{f_{P,K}(p_i, k)}. \quad (2)$$

The posterior probabilities $p(\mathbf{L} = \ell_i \mid (P, K) = (p_i, k))$ are derived from the model built during profiling.

Metrics for SCA. Model performance is evaluated using the rank function. Let $k^* \in \mathcal{K}$ be the true key used in profiling. Then,

$$\text{rank}(\hat{g}, \mathcal{D}_{\text{profiling}}, \mathcal{D}_{\text{attack}}, n) = |\{k \in \mathcal{K} \mid \mathbf{d}_{Na}[k] > \mathbf{d}_{Na}[k^*]\}|, \quad (3)$$

where $\mathbf{d}_{Na}[k]$ is derived from Equation 2. Rank is computed over n traces. A rank of 0 means k^* has the highest score; the worst rank is $|\mathcal{K}| - 1$. Average rank across chunks is termed guessing entropy [MPP16].

2.3 SHAP Values

SHAP (SHapley Additive exPlanations) values offer a method to interpret feature contributions in machine learning models, grounded in cooperative game theory [SREG⁺25].

For a model $f : \mathcal{X} \rightarrow \mathbb{R}$ and feature set $\mathcal{N}$, the SHAP value ϕ_i for feature i is defined by:

$$\phi_i = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!(|\mathcal{N}| - |S| - 1)!}{|\mathcal{N}|!} [f(S \cup \{i\}) - f(S)], \quad (4)$$

where S represents a subset of features. SHAP values satisfy additivity and fairness, making them suitable for analyzing contributions of time samples in side-channel traces.

2.4 Dataset and Experimental Setup

ASCAD Fixed (ASCAD-f). The ASCAD fixed dataset [BPS⁺18, BPS⁺17] is collected from an 8-bit AVR microcontroller executing a masked AES-128 implementation. Electromagnetic (EM) emissions are captured as side-channel traces during encryption. This dataset provides a profiling set comprising 50,000 traces and an attack set containing 10,000 traces. Each trace includes 700 time samples (i.e., features) extracted from a fixed temporal window. The same encryption key is used across both the profiling and attack sets. The primary objective of the attack is to recover the third key byte, which corresponds to the first masked intermediate value in the AES algorithm. In addition, this dataset offers parametrized desynchronization levels to assess algorithm robustness against jitter. In our work, we also utilize traces featuring 50 and 100 desynchronization settings [BPS⁺18].

Experimental setup. The experiments were conducted on a high-performance computing system running Ubuntu 22.04.3 LTS with a 6.5.0-44-generic Linux kernel. The system was equipped with dual Intel(R) Xeon(R) Silver 4216 CPUs, each with 16 cores and 32 threads, providing a total of 64 logical processors. The system had 256 GB of RAM. An NVIDIA RTX A4000 GPU with 16 GB of dedicated memory was utilized for GPU-accelerated computations.

3 Hardness Evaluation In SCA

Our objective in this section is to evaluate the hardness of the classification task of the SCA traces. To accomplish this, we first provide a formulation for NN-enabled SCA, then define three hardness metrics.

3.1 NN-enabled Profiled SCA

As elaborated in Section 2.2, the attacker’s objective is to recover the secret key k^* . To achieve this, she must determine which estimated probability distribution $\hat{g}_k$ (for each $k \in \mathcal{K}$) best corresponds to the observations in the attack set. Assuming reasonable conditions on the underlying distributions, the most effective approach is to utilize a *maximum likelihood* strategy.

This strategy involves evaluating the *likelihood function* $\mathbf{d}_{Na}[k]$ for every key candidate $k \in \mathcal{K}$ and selecting the key that maximizes this value, as formalized in Equation 2. Estimating $\Pr(\mathbf{L} \mid (P, K) = (p, k))$ for a pair $(p, k) \in \mathcal{Y}$, where $\mathcal{Y} \doteq \mathcal{P} \times \mathcal{K}$, is known as a prediction (or generation) problem. On the other hand, estimating $p((P, K) = (p, k) \mid \mathbf{L})$ is referred to as a classification problem. Bayes’ theorem connects these two formulations, enabling the more complex posterior to be derived from the generative model [BPS⁺18, LTR17].

NNs (NNs) have become increasingly relevant in SCA, as they offer a discriminative framework for handling classification tasks. In this context, NNs are used to directly estimate the posterior probabilities $\Pr((P, K) = (p, k) \mid \mathbf{L})$ without applying Bayes’ inversion [BN06]. The goal is to construct an approximation $\hat{f}_{\mathbf{L}, P}$ for the function $f_{\mathbf{L}, P} : (\ell, p) \mapsto \Pr((P, K) = (p, k) \mid \mathbf{L} = \ell)_{k \in \mathcal{K}}$.

Given an input trace ℓ and plaintext p , classification is performed by computing the output vector $\mathbf{y} = \hat{g}_{\mathbf{L}, P}(\ell, p)$ and selecting the most likely key candidate $\hat{k}$ via:

$$\hat{k} = \arg \max_{k \in \mathcal{K}} \mathbf{y}[k]. \quad (5)$$

When the attack is conducted using N_a trace-plaintext pairs $\{(\ell_i, p_i)\}_{i=1}^{N_a}$, the likelihood

score is aggregated using the maximum likelihood rule:

$$\mathbf{d}_{N_a}[k] = \prod_{i=1}^{N_a} \mathbf{y}_i[k], \quad (6)$$

where $\mathbf{y}_i$ is the prediction vector obtained by evaluating $\hat{f}_{\mathbf{L},P}$ on input (ℓ_i, p_i) .

Using the same setup, the model’s accuracy on a test dataset is given by:

$$\text{acc}(\hat{g}, \mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) = \frac{|\{(\ell_i, p_i, k^*) \in \mathcal{D}_{\text{test}} \mid k^* = \arg \max_{k \in \mathcal{K}} \mathbf{y}_i[k]\}|}{|\mathcal{D}_{\text{test}}|}, \quad (7)$$

where $\mathbf{y}_i$ is the output of the model $\hat{g}(\ell_i, p_i)$.

It is important to note that Equation 6 is essentially a practical reformulation of Equation 2, where the true posterior $\Pr((P, K) = (p_i, k) \mid \mathbf{L} = \ell_i)$ is approximated by $\mathbf{y}_i[k] = \hat{f}_{\mathbf{L},P}(\ell_i, p_i)[k]$ [BPS⁺18].

This approximation $\hat{g}_{\mathbf{L},P}$ is selected from a family of functions, typically known as models in the ML community. The specific form of this family is chosen by the evaluator (i.e., the adversary) based on problem characteristics. In SCA, initial studies focused on convolutional NNs (CNNs) with data augmentation to enhance model performance, even under countermeasure scenarios [BPS⁺18, PYW⁺17, WAGP20, ZBHV20].

Subsequent research revealed that multi-layer perceptrons (MLPs) can offer comparable or superior performance in certain settings [WPP22, AGF23]. Regardless of the neural architecture, the goal is to select an approximation $\hat{g}_{\mathbf{L},P}$ from a parameterized hypothesis space $G_\theta = \{g_\theta(x) \mid g_\theta : x \mapsto g_\theta(x, \theta)\}$, where $\theta \in \Theta \subseteq \mathcal{R}^q$, and q is the dimensionality of the parameter space [MDP20].

3.2 Hardness Metrics in SCA

As mentioned in Section 1, seen from various angles, existing literature has argued about the hardness of SCA, especially under a non-worst case scenario (i.e., the adversary does not know the values of the random nonces [MCLS23]). It is noteworthy that we do not particularly refer to the difficulty of applying NN-based techniques such as hyperparameter tuning, lack of relevant metrics, etc. [PPM⁺21], but the challenges the NN face when deciding about the correct key. Although this decision is indirectly related to the difficulties mentioned above, it is not solely influenced by them. [MCLS23, RHMM25] have linked the hardness of SCA to the Gradient Descent (GD)-based optimization algorithm. It is suggested that when targeting leakage induced by masking, the GD algorithm initially becomes stalled on a plateau corresponding to the full randomness. Nevertheless, the problem of distinguishing between the correct key and others remains the same even for algorithms not applying GD, e.g., InfoNEAT [AGF23]. While InfoNEAT has demonstrated competitive results, it still requires tens of attack traces to achieve rank 0.

A more general argument has been given in [MO23], where it is hypothesized that the main difficulty for any NN is to find the leaking trace points, ones with strong leakage. Such an argument is in line with studies devoted to finding point-of-interest in leakage traces, e.g., [PWP22, RBFA21, YPB24]. Pre-processing techniques to determine point-of-interest in traces and feed them to the NN could be helpful in this regard; nonetheless, when combined with NNs’ capabilities to learn the leakage distribution conditioned on the point-of-interest, the attack performance would be boosted.

Overfitting and being over-confident have also been identified as contributing factors [MO23, NG24]. In fact, when the correct key has a high rank during the attack, it indicates that the NN is overconfident. Such a network typically produces a softmax output distribution in which some classes have a high probability while the rest have values close to zero [GPSW17]. Overfitting also results in the correct key’s high rank due to the network’s

incapability to generalize to unseen traces. Recent studies have devised a mechanism to tackle this by incorporating the information on labels in training. [WPP21] has attempted to leverage the correlation between label distance and embedding distance. The training metric proposed in [WWK⁺23] has shared the same goal to improve generalization by employing the correlation between the key distribution and the key guessing vector of all key guesses.

The evolving line of research dedicated to understanding the hardness of NN-assisted SCA raises the question of how similar challenges have been addressed within the ML field.

3.3 Hardness from ML Perspective

This section aims to bridge the gap between our understanding of side-channel attack hardness and the hardness metrics devised in the context of NN computation. As discussed above, hardness of SCA can be broadly categorized into hardness of training (profiling) [WWK⁺23, WPP21, MO23, MCLS23] and/or hardness of testing (attack) [NG24]. Seen through the lens of NN computation, hardness of training refers to how difficult it is for the model to learn from specific training examples, in particular, learning the similarities between a set of inputs (traces) is hard. Although it might indirectly lead to slow convergence, high error, and frequent misclassification, SCA mainly has to deal with situations where a NN assigns relatively low probability to its predictions [BFT17]. This means that the output of the softmax layer does not strongly favor the correct class, i.e., high rank and unlikely distinguishable keys. On the other hand, the hardness of testing signifies that the model finds it difficult to classify some inputs correctly. These examples often lie near decision boundaries or in regions of low confidence in the learned feature space. This behavior is also linked to low confidence predictions [MWX⁺22], specifically, the network outputs a softmax distribution with no dominant class in SCA.

To put the above discussion in context, we should first define the notion of hardness formally. A deep NN defines a non-linear mapping function f_θ , made by a set of parameters θ (representing the network’s weights). This function transforms inputs from a measurable space $\mathcal{X}$ into outputs within a measurable space $\mathcal{Y}$ as $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ with $f_\theta(x) = y$. Within the context of supervised learning, we are given a finite training dataset $\mathcal{D}$, which is a subset of the input-output domain $\mathcal{D} = \mathcal{X} \times \mathcal{Y}$. This dataset consists of N input-target pairs and is denoted as: $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$. Once trained on $\mathcal{D}$, the NN can generalize to unseen samples. For any new input $x^* \in \mathcal{X}$, it predicts a corresponding output y^* using the learned function: $f_\theta(x^*) = y^*$. The central question we aim to address is: how difficult is it to classify a new, unseen sample x^* in comparison to the already seen samples? To quantify this, we define a function $h(x^*) : \mathcal{X} \rightarrow \mathcal{R}$ such that if $h(x_1^*) > h(x_2^*)$, x_1^* is more difficult to classify than x_2^* .

Next, we explore three hardness metrics suitable in the context of SCA, borrowed from the NN computation field.

Query hardness (QH). In NN computation, a “query” represents a new sample that the NN has yet to see. Query hardness (QH) quantifies the semantic dissimilarity between a query and the set of samples the network has observed (hereafter called “support set”). QH [YZSH20] provides a way to measure this difficulty of classifying a new sample, which is essential for analyzing model performance across varying degrees of classification difficulty. QH is calculated using the Equation 8.

$$h = \log \left(\frac{1-s}{s} \right); s = \frac{\exp(\langle r, p_{c=k^*} \rangle)}{\sum_c \exp(\langle r, p_c \rangle)}. \quad (8)$$

This equation computes a similarity score s , which measures how close the query’s is to the support set through comparing the cosine similarity between the query’s prediction logits (r) and the average prediction logits of class c in the support set (p_c). Moreover, $\langle \cdot \rangle$

refers to the cosine similarity, which measures how similar two vectors are in terms of their orientation. r represents the logits (raw output values before activation) for the query, and p_c represents the average logits for each class in the support set. The term k^* denotes the ground truth label for the query, i.e., the correct key. This equation transforms the similarity score into a measure of hardness, where higher values of h indicate that the classification task is more difficult.

The importance of QH lies in its ability to assess how challenging a particular classification task is for the model. For example, a model might perform well on easy queries but poorly on harder ones. By examining the hardness scores, we can identify which types of queries the model struggles with and work towards improving its performance in terms of generalization.

When it comes to SCA, QH matches the analysis’s objective well. This is thanks to the fact that, unlike traditional hardness metrics relying on model confidence scores or loss values [JKGG18, SGG16], QH evaluates the *semantic* dissimilarity between a query sample and its corresponding support set. When training NNs for profiled SCA, models benefit from identifying semantically similar traces, which enables the model to generalize better to unseen attack traces, even if those traces appear different in form due to countermeasures like masking or desynchronization.

Log odds (LO). Log odds (LO) is a metric designed to assess the average difficulty of a batch by evaluating how likely it is for unseen samples to be misclassified [DCRS19]. For this purpose, it measures the average disparity between the softmax distributions of each class and other classes, providing insight into the confidence of the model’s predictions. Concretely, given a support set D_s and a set of unseen samples D_q , the LO is computed as:

$$\Omega_\phi(D_q; D_s) = \frac{1}{N_q} \sum_{(x,y) \in D_q} \log \left(\frac{1 - p(y | x)}{p(y | x)} \right), \quad (9)$$

where $p(y | x)$ is the softmax probability distribution of the true label y for the input x , derived from a softmax function applied to the logits $z_y = w\phi(x)$. Here, w represents the weight matrix computed from the support set, and $\phi(x)$ denotes the ℓ_2 -normalized logits generated by the NN.

The LO score Ω_ϕ reflects the model’s confidence in distinguishing between the true label and other classes. A higher value of LO indicates that the model is less confident about the classification and more likely to misclassify the unseen samples. Conversely, a lower value suggests that the model is more certain about the predictions. In other words, this score serves as an indicator of how well the features in the unseen set can be linked to the features from the support set.

Application of LO in SCA can be justified by LO’s intrinsic characteristics, namely, the ability to quantify how well a model distinguishes between different classes. In this sense, a low LO for the correct key indicates that either (1) the traces are noisy, (2) labels are closely confusable, and/or (3) the NN has low discriminative power. Compared to QH, LO offers a higher degree of freedom since it does not require computing the cosine similarity. Furthermore, QH can be employed for NNs with activation functions from the ReLU family [YZSH20]. Nonetheless, LO gives a high-level picture of the softmax probability distribution since it represents the average over a subset of unseen samples. A more fine-grained view of the softmax probability distribution can be provided by entropy, as explained below.

Matrix-based Rényi Entropy (MRE) Our observation points out that the probability mass is spread over the output classes due to multiple factors enumerated above in SCA. This means that if all classes have low and (somewhat) similar probabilities, a high rank of the correct key is achieved. To capture this effect, entropy can be proposed to quantify how probability mass is distributed among all classes.

Previous studies employing entropy in SCA have described why MRE is an appropriate choice, cf. [NG25]. First, SCA often involves high-dimensional data and large datasets, for which matrix-based formulations offer computational advantages by enabling more efficient processing and reduced complexity. Second, and more critically, information-theoretic measures such as entropy inherently depend on the underlying probability distribution $H(y|x)$. When the true distribution is unknown and only a finite sample is available, these measures must be estimated [GRP14]. While straightforward and sometimes computationally convenient, off-the-shelf functions for computing entropy are highly sensitive to the accuracy of the estimated data distribution, which can be challenging in the presence of noise or limited sample sizes.

By adopting a generalized form of Shannon entropy, specifically, Rényi entropy, information-theoretic quantities can be formulated as functionals over normalized positive definite matrices. These matrices are constructed by evaluating a positive definite kernel function across all pairs of data samples [GRP14, YWJP20]. This kernel-based approach implicitly maps the data into a reproducing kernel Hilbert space (RKHS), enabling entropy computation without requiring an explicit estimation of the underlying probability distribution.

(cf. [YP19]) Consider a dataset consisting of n samples $\{x_1^i, x_2^i, \dots, x_k^i\}_{i=1}^n$, where each sample comprises k measurements (with $k \geq 2$). Define a set of positive definite kernels $\kappa_1 : \mathcal{X}_1 \times \mathcal{X}_1 \rightarrow \mathcal{R}, \dots, \kappa_k : \mathcal{X}_k \times \mathcal{X}_k \rightarrow \mathcal{R}$ such that $X_z = \{x_z^1, \dots, x_z^n\}$ for $1 \leq z \leq k$.

The kernels $\kappa_1, \dots, \kappa_k$ are assumed to be real-valued, positive definite, and infinitely divisible [Bha06, YP19]. The Rényi entropy of order α for the joint distribution of k variables is defined by:

$$\mathbf{J}_\alpha(\mathbf{X}_1, \dots, \mathbf{X}_k) = \mathbf{S}_\alpha \left(\frac{\mathbf{A}_1 \odot \dots \odot \mathbf{A}_k}{\text{tr}(\mathbf{A}_1 \odot \dots \odot \mathbf{A}_k)} \right), \quad (10)$$

where the matrix elements are given by $(\mathbf{A}_1)_{ij} = \kappa_1(x_1^i, x_1^j), \dots, (\mathbf{A}_k)_{ij} = \kappa_k(x_k^i, x_k^j)$. Here, $\odot$ denotes the Hadamard (element-wise) product, and $\text{tr}(\cdot)$ represents the trace of a matrix.

The function $\mathbf{S}_\alpha(\cdot)$ is then defined as:

$$\mathbf{S}_\alpha(\mathbf{A}) = \frac{1}{1-\alpha} \log_2(\text{tr}(\mathbf{A}^\alpha)) = \frac{1}{1-\alpha} \log_2 \left(\sum_{i=1}^n \lambda_i(\mathbf{A}^\alpha) \right), \quad (11)$$

where $\lambda_i(\mathbf{A})$ denotes the i^{th} eigenvalue of the matrix $\mathbf{A}$.

The connection between the matrix $\mathbf{A}$ in Equation (11) and the Gram matrix $\mathbf{K}$ is given by the normalization $\mathbf{A}_{ij} = \frac{\mathbf{K}_{ij}}{n\sqrt{\mathbf{K}_{ii}\mathbf{K}_{jj}}}$.

To approximate Shannon entropy using this framework, one can evaluate the Rényi entropy with $\alpha = 1.01$, as suggested in previous works [GP13, GRP14]. This provides a stable numerical approximation of information quantities derived from the data without density estimation.

Summary. This section provides insight into how attack hardness has been seen from two perspectives, namely SCA and neural network computation. It also summarizes three hardness metrics borrowed from the latter field, which can help understand the attack difficulty at different levels of granularity and with various degrees of freedom. Next, we explore how different factors affecting the softmax output distribution contribute to the hardness of an attack.

4 Do Labels Contribute to the SCA Hardness?

In profiled SCA, each trace is labeled according to the target intermediate value and leakage model. The goal is for the model to learn the semantic meaning of that label from

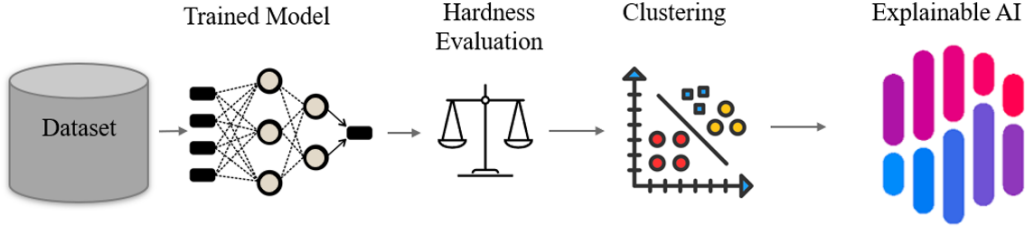


Figure 1: Using XAI to understand the influence of different factors on the hardness of SCA.

the leakage. Generally speaking, labels play a significant role in determining the hardness of training/testing, i.e., how difficult it is for the model to correctly classify certain unseen examples. First, fuzzy boundaries between labels, i.e., classes with overlapping or highly similar feature representations, can confuse the NN. This can be observed in SCA when using the Hamming weight leakage model, where different sub-keys (classes) are combined.

Second, underrepresented labels during training often lead to poor generalization at test time. This has been well understood in SCA [PHJ⁺19a]. Third, the semantic complexity of labels makes it harder for the NN to generalize. This notion represents how conceptually diverse or rich in meaning a class label is. Label representation can significantly impact the perceived semantic complexity of labels, i.e., how ambiguously a label’s meaning is encoded. When applying one-hot encoding as a typical method in SCA, each class is treated as equally distinct, with no semantic similarity between labels. Hierarchical label representation as applied in [AGF23] decomposes complex labels into simpler, interpretable sub-labels. As a result, the NN can handle generalization more effectively.

On the other hand, if the label similarities are encompassed in training, training/testing hardness can be reduced. This has received attention recently in SCA literature. [WPP21, WWK⁺23] have proposed two techniques. While [WWK⁺23] has transferred the one-hot encoded labels to Gaussian distributed ones, [WPP21] has optimized the distance metric (Euclidean distance) to enforce training based on both embedding and label distances. In the former case, the codes are represented in integer format, while in the latter, labels are one-hot encoded. This, of course, impairs the effectiveness of their approaches. Next, we deploy a strategy relying on XAI to understand to what extent the approaches in [WPP21, WWK⁺23] could reduce hardness.

4.1 Methodology for Explaining Hardness

For the results of this section, we use the model presented in [WAGP20] that was proposed for the ASCAD dataset [BPS⁺18] with no desynchronization. This model consists of an average pooling layer with size 350 and a flatten layer, two fully connected layers with 10 neurons each, and a softmax layer with 256 outputs.

As depicted in figure 4.1, we first feed the trained models with the traces and then output probabilities and logits, which are used to calculate hardness metrics. In our case studies, we calculate the hardness value associated with each label for both profiling and attack datasets separately by using exactly 4000 traces for each case. We introduced three hardness metrics in Section 3, for QH and LO, the hardness value assigned to a label is computed as the average hardness of all traces with that label, whereas for MRE, it is calculated as the sum of the hardness values across all traces sharing the same label to incorporate the upper bound of uncertainty for the traces [Cov99]; hence, MRE reflects the maximum entropy conditioned on each class. Next, we apply standardization to the hardness values of each dataset, as the SHAP analysis is sensitive to the numerical value of the target feature, and we want to have a fair comparison between the metrics, and the

standardization helps to avoid any numerical issues.

In the second step, the optimum number of clusters is found using unsupervised ML methods [HSW01] to find the classes that are located in the same clusters. Next, we use three different measures considered in the SCA-related literature to examine the influence of labels on hardness. First, we ask the question of whether there is a significant difference between label bits when comparing the hardness of profiling/attack. Afterward, we turn to the question of whether a smaller label distance may indicate similar leakage, cf. [WPP21], and consequently, more ambiguous decisions made by the NN and higher hardness value. Finally, we shift our focus to investigating if label correlation can reduce the ambiguity and make profiling/attack easier as suggested in [WWK⁺23]. In our analyses, we explain relationships between hardness and label-related metrics based on the SHAP values [SREG⁺25].

What is the statistical distribution of the hardness values? As can be seen in Figure 1, the core idea behind our approach is to cluster similarly hard classes together. This is the basis to examine whether their cluster-forming behavior can be related to (1) labels bits, (2) smaller label distance [WPP21], and (3) correlation between label and softmax-output distributions [WWK⁺23]. In the first step, we need to assume a distribution for the hardness values, as we need to understand which clustering algorithm and which inter-distribution distance metric should be employed. Given that the traces can be multivariate (and/or multi-model) Gaussian-distributed [LPMS18, LM18, RBAF24, MMOS16, BGH⁺17, GLRP06, RBFA21, DCGRP19, Riv08, SMY09] and the neural network is smooth and continuous [HSW89, GBC16], we expect the logits and softmax outputs to preserve class structure to some extent. Applying the mappings in Equations 8, 9, 11, we aim to understand how smooth and Gaussian-like the hardness values could be. If the resulting hardness values are well-clustered in the simplex space, k-means [Seb09] could serve as a fast and reasonable clustering method. To evaluate the validity of the assumption, we fit a Gaussian distribution to the hardness values and apply the quantile-quantile (QQ) plot [GW68].

QQ-plots. A quantile-quantile plot (Q-Q plot) is a visualization tool used to assess whether a set of sample data follows a specified distribution. To draw the plot, the sample values are sorted in ascending order and plotted against the corresponding theoretical quantiles of the reference distribution, i.e., the Gaussian distribution in our case. The y-axis represents the ordered sample values, while the x-axis shows the theoretical quantiles. If the points in the plot align closely with a straight line, this suggests that the sample data likely follows the specified distribution. Concretely, for a sample of size n , the Q-Q plot uses n quantiles, plotting the i -th ordered value against the quantile at $(i - 0.5)/n$ of the Gaussian distribution [Cha18, GW68]. Using the experimental setup as described in Section 2.4 we obtain hardness values according to Equations 8, 9, 11. Figure 2 shows the QQ-plots of the hardness values. As seen in this figure, hardness values shown by the blue markers align with the diagonal reference, which means the assumption of the normal distribution is correct.

Which classes are similarly hard? To analyze which classes are equally hard, we use k-means [Seb09] to cluster values as mentioned before. One critical aspect of every classification/clustering algorithm is to select the appropriate distance metric to measure the closeness of data points to each other. Not only does the chosen distance metric significantly impact how well algorithms perform, but also the choice of metric can influence the explainability of results. Euclidean distance is the theoretically justified distance metric for k-means clustering. This is due to the fact that k-means minimizes within-cluster variance, corresponding directly to Euclidean distance in a least-squares sense [Mac67, Jai10]. Moreover, as k-means minimizes the sum of squared distances to the cluster centroids, we take the square Euclidean distance into account. This greatly simplifies the optimization.

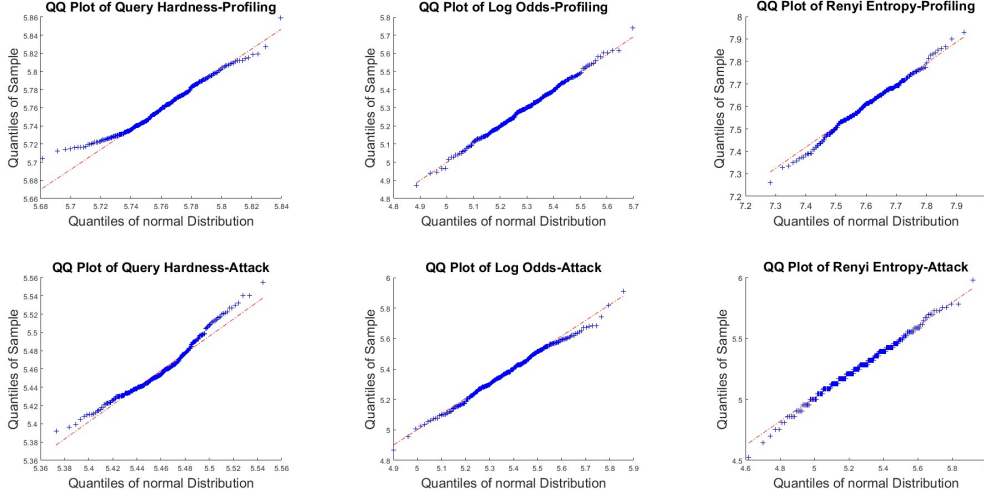


Figure 2: QQ-plots for three different hardness metrics while using attack and profiling traces. In these plots, the x-axis is for the quantiles of the normal distribution, and the y-axis is for the quantiles of the hardness values. Blue points roughly aligning with the diagonal reference line confirm that these values follow a normal distribution.

Another important aspect of clustering is determining the number of clusters. To determine the optimal number of clusters, k-means relies on heuristic methods such as the Elbow method [Tho53] and the Silhouette method [Rou87]. The Elbow method considers the trade-off between the number of clusters and the within-cluster variance (so-called, inertia), identifying the point where adding more clusters is not beneficial to the algorithm performance. In contrast, the Silhouette method quantitatively assesses the quality of clustering by measuring how well-separated the clusters are. Using a score that ranges from -1 to 1, the Silhouette method suggested the optimal number of clusters corresponds to the highest average Silhouette score. While the Elbow method offers an intuitive guide, the Silhouette method provides a more data-driven measure of cluster validity. Nevertheless, in our experiments, we have observed that when clustering the hardness values computed via MRE, the clusters overlap; hence, we applied the Elbow technique in that case.

After finding the optimal cluster number, we apply k-means clustering to find classes with similar hardness. Next, we design three algorithms to see what makes the classes have the same hardness level.

4.2 Do Label Bits Influence the Hardness?

First and foremost, we stress that labels fed to NN classifiers are *representations* of the intermediate values. Irrespective of the leakage model, labels are often given to the NN in one-hot coded format. This encoding disconnects the labels and the traces. The intermediate values indeed have a strong connection to the leakage; however, one-hot coded labels may not reflect this relationship. Therefore, we aim to answer the question of “do the label bits impact the hardness of the class?” To this end, we turn to the k-means’ sister algorithm, namely k-nearest neighbor (KNN) regression. The key insight into this is that after clustering the hardness values, different labels are grouped together. We attempt to analyze whether these groups also capture any relationship with label bits. For instance, consider a cluster composed of class “0” $(1, 0, \dots, 0, 0)$ and class “5” $(0, 1, 0, \dots, 0, 0)$. Can we conclude that the most significant bit (MSB) and the second most significant bit (SMSB) are not influential in the hardness of these classes? KNN regressor finds the k

closest labels and attempts to relate them to their hardness level. Afterward, we can use the trained KNN regressor as a function to be given to a SHAP explainer 2.3 to explain the hardness against label bits mapping. As defined in section 2.3, SHAP values of a feature are computed considering all possible subsets of other features' contributions, which is aligned with the multivariate nature of the SCA datasets. For example, when we calculate the contribution of the first bit, we can just flip that bit and see how much the decision of the model is changed without considering the influence of the other bits. On the other hand, SHAP values consider all possible subsets of other features and take the average value of the changes.

Results. Figure 3 shows the results when attempting to explain the hardness based on the label bits. In these plots, the x-axis represents the bits of the labels, where bits 0 and 7 are the most and the least significant bits (MSB and LSB), respectively. While the first row is for the profiling dataset, and the second row is for the attack dataset. Each point represents the SHAP value corresponding to a label bit. The SHAP value represents the average changes of the hardness value when a label bit flips, while considering all possible subsets of other bits. Positive values in the figure indicate that the contribution to the regression model was positive, and negative values indicate a negative contribution. For example, if the SHAP value assigned to a bit is positive for a trace, it means that this bit pushes the hardness above the mean value.

Starting with QH, bit 0 has the most contribution for the profiling traces, while other bits contribute almost the same. In the attack case, the contributions of bits 1 and 2 are more dominant. For the LO metric, bits 0 and 1 are the most important bits in the profiling dataset, while bit 2 has the most contribution in the attack dataset. Bit 1 and bit 2 contribute the most in the MRE metric profiling dataset; in the attack dataset, bit 0 and bit 2 contribute the most. The most important message given by these sub-figures is that the distribution of SHAP values per bit is symmetrical, with the mean equal to zero. A symmetrical distribution of SHAP values suggests that each bit has a balanced influence on the hardness, i.e., pushing both positively and negatively with similar frequency and magnitude. For some bits, we can also see that the SHAP values are also tightly clustered around zero, which may indicate that the feature group has limited overall impact. Conversely, a wide but symmetrical distribution could reflect that the group is informative, yet its influence varies across different samples.

4.3 Does Minimum Distance Between Labels Influence the Hardness?

The labels in SCA are derived from intermediate values. Leakage models can also be applied to the labels when analyzing the leakage. [WPP21] has conjectured that a smaller distance between labels may reflect more similar leakage traces, indicating a smaller distance in feature space. Here, we aim to shed light on two important aspects of this: (1) which distance metric should be used when comparing labels? And (2) if it can be demonstrated that the conjecture is true. Intuitively, as discussed before, the labels are representations of the intermediate values, although the connection between the intermediate values and leakage traces is very much affected by this representation.

Distance metric. [WPP21] has considered normalized Euclidean distance between labels ranging from zero to one (cf. Equation 3 in [WPP21]). According to the code accompanying [AIS21], integer labels have been used in their study. In the context of SCA, integer labels represent categories, not numerical quantities. Treating them as integers implies ordinal or metric structure, which often does not exist [JWH⁺13, Gér22]. In fact, [Gér22] emphasizes that categorical variables should be one-hot encoded instead of integer encoding. This is because assigning numerical values to categories imposes an unjustified ordinal relationship. Employing normalized Euclidean distance is not recommended since it inherently treats numbers as lying on a continuous space, which is not valid for sub-keys (binary values) represented as categories. Compared to normalized

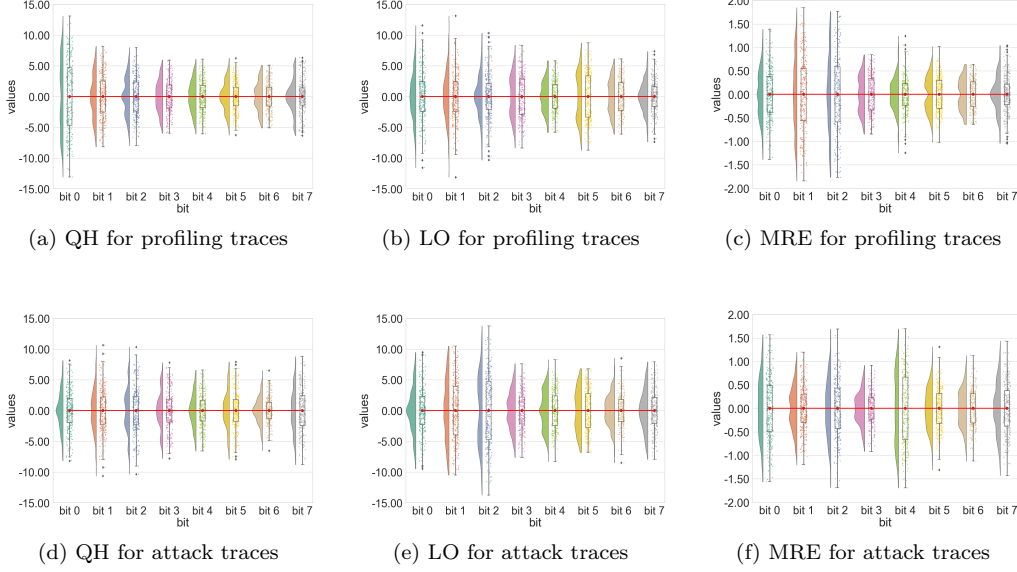


Figure 3: The distribution of the SHAP values for the contribution of the label bits to the hardness values. For similarly hard classes, an attempt is made to explain their hardness based on the label bits. The first row corresponds to the profiling traces, while the second row is for the attack traces. In both cases, 5000 traces of the ASCAD dataset [BPS⁺17] are used.

Euclidean distance applied to integer labels, Jaccard distance applied to the one-hot encoded labels provides a more principled and interpretable measure. In this case, Jaccard distance respects the class labels’ categorical, discrete nature. In doing so, it avoids introducing artificial ordinal structure and directly reflects whether two labels are close [TSK16]. The Jaccard distance is a widely used metric for quantifying the dissimilarity between two binary vectors, defined as

$$d_{\text{Jaccard}}(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}, \quad (12)$$

where A and B are binary vectors, $|A \cap B|$ denotes the number of features where both A and B have a value of 1, and $|A \cup B|$ represents the total number of features where either A or B is 1. The Jaccard distance ranges from 0 to 1, with 0 indicating identical vectors and 1 indicating complete dissimilarity. This metric is particularly well-suited for binary or categorical data, focusing exclusively on the proportion of shared non-zero features while ignoring zero values. In our study, the Jaccard distance is employed to measure the dissimilarity between the binary representations of class labels within clusters, facilitating a meaningful comparison of structural differences in the label space.

How to examine the influence of the minimum distance between labels on hardness? As described above, the suitable distance metric is the Jaccard distance applied to binary labels; however, to closely study the conjecture in [WPP21], we also apply the Euclidean distance. As in Section 4.2, we apply the KNN regressor, where the labels and minimum distance between similarly hard classes are fed to the regressor. Again, LO, QH, and MRE hardness metrics are used to compute the hardness value and cluster similarly hard classes together. Afterward, the minimum distance between cluster members is considered, where both the Euclidean and Jaccard distance metrics are investigated. For example, if two classes are in two different clusters, but the minimum distance is the same

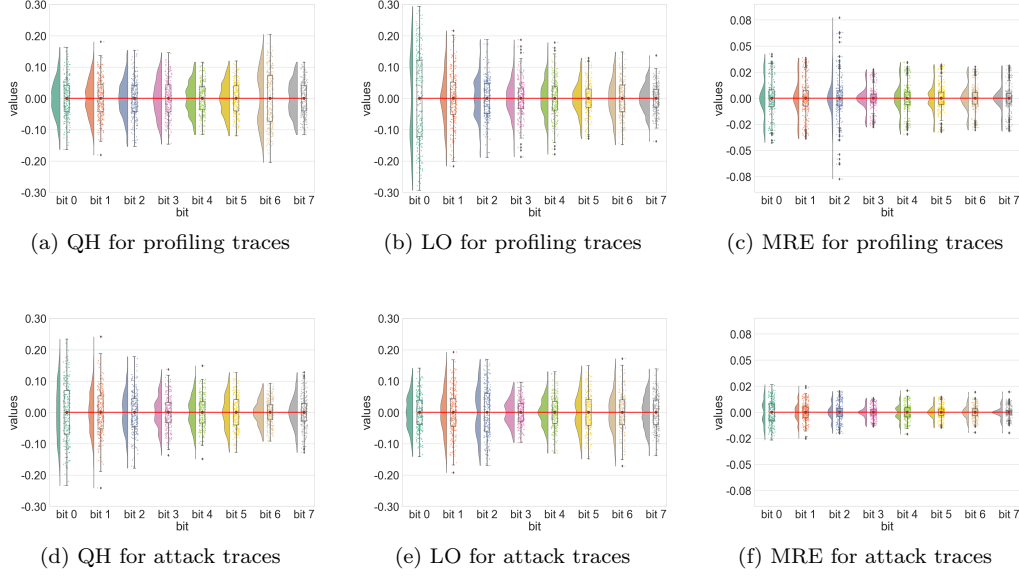


Figure 4: The distribution of the SHAP values for the contribution of the label bits to the hardness values. The minimum distance between labels is calculated for similarly hard classes using the Euclidean distance metric. The first row corresponds to the profiling traces, while the second row is for the attack traces. In both cases, 5000 traces of the ASCAD dataset [BPS⁺17] are used.

in those clusters, then the labels will have the same target value in the regressor.

Results. Figure 4 shows the results for the scenario, where the minimum distance between similarly hard classes is measured using the Euclidean distance. The first and second rows include the results for profiling and attack traces, respectively. In QH metric plots, bits make almost the same contribution to the profiling dataset, suggesting a balanced contribution of each label bit for the profiling dataset. However, bits 0 and 1 have slightly more contributions in the attack dataset. For these two bits, the majority of the distribution of the SHAP values is around zero. While bit 1 has more contribution in the attack dataset in the hardness values, bit 0 has a higher contribution in the profiling dataset in the eyes of the LO metric. MRE plots are a slightly different from the other metrics. For the profiling dataset, we can see that a couple of points of bit 2 have higher SHAP values. The first two bits of the labels in the attack dataset have subtly higher variance and more importance in the SHAP analysis. For MRE plots, we can see that the ranges of the SHAP values of the attack dataset are smaller than those of the profiling dataset. Our analysis shows that this comes from the lower number of clusters that group the similarly hard classes in the attack dataset compared to the profiling dataset.

When we use the Jaccard distance to measure the minimum distance between labels, as shown in figure 5, the range of the SHAP values decreases compared to figure 4. We can see that the SHAP values of bits 0 and 1 are higher in both datasets for the QH metric. While bit 0 has a greater contribution to the profiling dataset. In LO metric plots, bit 0 has the highest SHAP values for the profiling plots, while bits 1 and 2 are dominant in the attack dataset. Here, changing to the Jaccard metric did not affect bit 2, contributing the most for the profiling dataset in the eyes of the MRE metric. Again, we see that the ranges of the SHAP values of the attack dataset are less than those of the profiling dataset. For the results presented in Figures 4 and 5, we can see that the distribution of the SHAP values is centered around zero in all cases. This indicates that the contribution of the

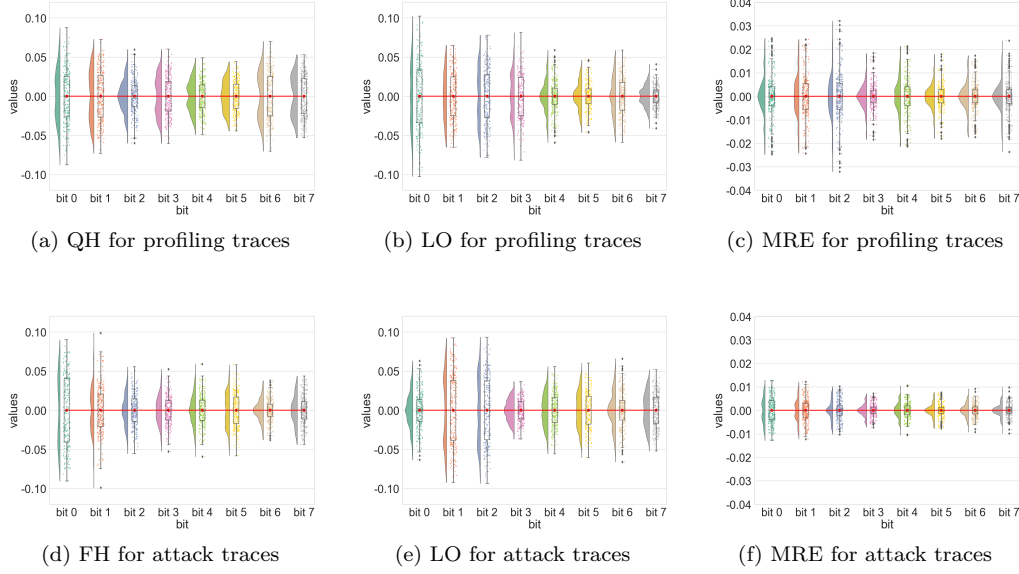


Figure 5: The distribution of the SHAP values for the contribution of the label bits to the hardness values. The minimum distance between labels is calculated for similarly hard classes using the Jaccard distance metric. The first row corresponds to the profiling traces, while the second row is for the attack traces. In both cases, 5000 traces of the ASCAD dataset [BPS⁺17] are used.

labels’ bits to the classification task’s hardness is subtle. Furthermore, compared to the results in Figure 3, the SHAP values are more narrowly concentrated around zero, implying that the label bits have minimal overall importance. Again, as discussed in Section 4.2, the label bits have a balanced effect on the minimum distance of similarly hard classes due to the symmetrical distribution of SHAP values.

4.4 Does Correlation Between Labels Influence the Hardness?

The notion of label correlation (LC) in SCA has been introduced in [WWK⁺23] to incorporate the relationship between the output probability of the model and the ground truth labels. LC is defined as

$$LC = \rho(d_{SE}, \text{Rank}(\mathbf{y})), \quad (13)$$

where $\text{Rank}(\mathbf{y})$ denotes the sorted key guessing vector, which is the vector of probabilities for all key candidates from the output of NN’s softmax. d_{SE} is the distance between key candidates and the correct key, measured in square Euclidean distance. When the correct key is unknown (black-box), the most likely key serves as the reference key, cf. [WWK⁺23]. The square Euclidean distance metric is used following the label smoothing applied so that labels follow a Gaussian distribution. Hence, we also use this distance metric in our analyses.

How to examine the influence of the label correlation on hardness? As before, we group the classes based on their hardness and attempt to explain what makes them similarly hard. Per cluster, we need a representative value to observe the contribution of the LC to the hardness. For this purpose, we apply the average LC, i.e., the Pearson correlation averaged over similarly hard classes within a cluster. One effective approach is

to use Fisher’s z-transformation [BW95, CDB98]:

$$\bar{\rho} = \tanh \left(\frac{1}{K} \sum_{j=1}^K \tanh^{-1}(\rho_j) \right) \quad (14)$$

We take this approach as it reduces skewness in the distribution, hence providing a representative statistical measure.

The flow of our approach is comparable to that described for label bits and the minimum distance between labels. However, some changes should have been applied to follow the core idea in [WWK⁺23]. First, the algorithm loads the softmax-output distribution and the actual class labels. The output distribution is fitted to a Gaussian distribution as suggested in [WWK⁺23], where the fitted values are used to compute the distance d_{SE} in square Euclidean distance. Sorting these distances by proximity is followed by calculating Spearman correlation between the ranked prediction distances ($\text{Rank}(\mathbf{y})$) and d_{SE} , indicating the degree of alignment between them. Following the initial LC calculations (Equation 13), we perform clustering on class hardness values. For each cluster, we compute an average LC score (Equation 14), tracking the correlation allows us to calculate the contribution of the LC to the hardness metrics using SHAP values.

Results. Figure 6 shows the results of the SHAP analysis using hardness clustering based on label correlation. For the LO metric, the first two bits show more contribution in both datasets. While bit 2 has a higher range of the SHAP values in the attack dataset, all the bits (except for bits 5 and 6) contribute the same for the profiling traces in QH plots. The first three MSBs have the most contribution to the profiling dataset of the MRE metric. When we focus on the attack dataset, we can see that all the bits contribute almost the same.

Analyzing the ranges of the plots in figure 6, we conclude that by using the label correlation as the representative of the hardness clusters, we could not find any significant contribution from the bits of the label in the hardness values. Although some bits have slightly more contribution than others, this was not consistent for all metrics and even for those bits, the majority of the SHAP values are around zero. As also explained in Sections 4.2-4.3, a low overall contribution of the label bits and label correlation to the hardness can be observed as the SHAP values are tightly concentrated around zero.

Summary. Based on the experiments conducted in the Results and Methodology sections, we observe no significant correlation between the labels of the ASCAD dataset and the hardness of the classification task. The analysis included detailed evaluations of various classification metrics such as log-odds error, entropy, and query hardness, where the hardness of the task was assessed using an XAI tool, namely SHAP explainers. The findings consistently indicate that the inherent difficulty in classifying specific labels is not systematically tied to the labels themselves. This suggests that factors unrelated to the label representation might contribute to the observed hardness, reaffirming that the hardness of the task is less label-independent.

5 Which Time Samples Contribute to Attack Hardness?

In Section 4, we have investigated whether labels could contribute to the hardness of SCA. Our analysis found no statistically significant evidence to support the existence of such a relationship. Now, we turn to the question of whether trace features (time samples) lead to the hardness of SCA. Here, we aim to explore that, given an unseen sample $x^* = [x[0], x[1], \dots, x[t]]$, what is the contribution of each time sample (or a subset of them) to the hardness value $h(x^*)$. In order to answer this question, we need a function that maps the unseen traces x^* to the hardness values, so that we can use that function for SHAP analysis. The reason that we use this methodology is twofold. First, a function

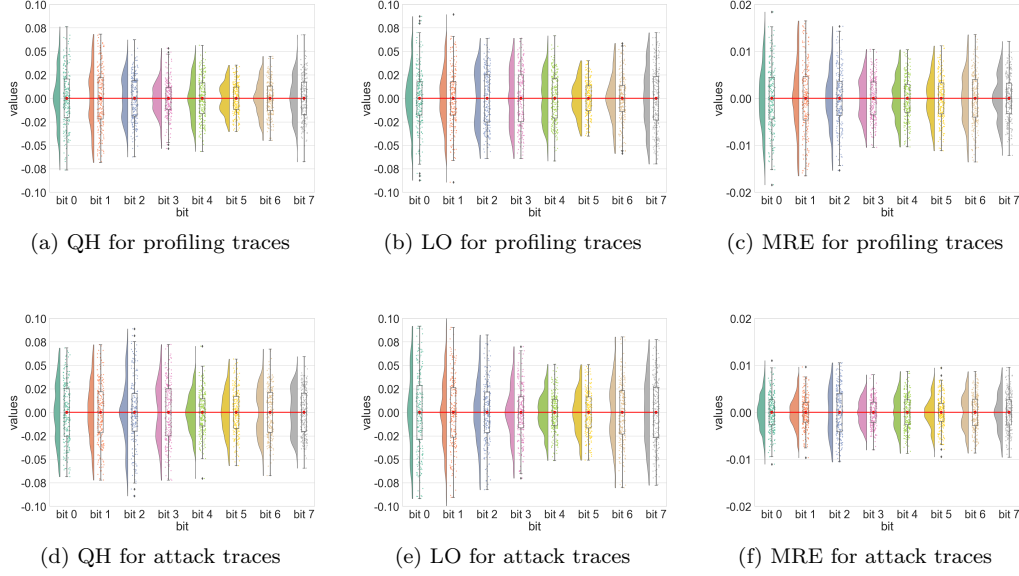


Figure 6: The distribution of the SHAP values for the contribution of the label bits to the hardness values. Label correlation is calculated for similarly hard classes. The first row corresponds to the profiling traces, while the second row is for the attack traces. In both cases, 5000 traces of the ASCAD dataset [BPS⁺17] are used.

that directly maps the unseen traces to the hardness values provides more accurate results than pipelining the classification model $f_\theta(x^*)$ with the hardness function h . Second, we prefer SHAP analysis over gradient computation, as in SHAP values, the influence of the time instances on each other is also calculated. Moving further with this methodology, our choice is ridge regression [HK70].

Ridge regression is particularly well-suited for this task due to its ability to handle multi-collinearity among features, which is common in high-dimensional datasets like those composed of side-channel traces with many time samples. By regularizing the regression coefficients, ridge regression minimizes overfitting and ensures a more stable solution than ordinary linear regression [HTFF09]. This is especially important in our context, as the traces contain highly correlated time samples. Therefore, as the first step, we train a ridge regression model on the traces and hardness values for SHAP analysis. In the second step, we calculate the SHAP values. By grouping the traces into four categories. Each category corresponds to a different target variable as defined based on their SNR in [BPS⁺18], see Figure 7. We further explore the distribution of contributions.

Results. In Figure 8, the SHAP values are presented. The positive value (blue color) in these plots indicates that the corresponding instances for these classes contribute positively to the hardness value (making classification harder). In contrast, negative (red color) values mean that instances reduce the hardness of the classification. Our background information about the ASCAD dataset [BPS⁺18, BPS⁺17], it is known that the original traces (each consisting of 100,000 samples) are truncated to retain only the 700 samples within the interval [45,400–46,100]. This interval contains relevant information for the two S-box-related pairs: $(Sbox(p[3] \oplus k[3]) \oplus r[3], r[3])$ and $(Sbox(p[3] \oplus k[3]) \oplus rout, rout)$, which are denoted in the plots as MOL, OML, MO, and CM, respectively. Therefore, we expect that per class, (virtually all) time samples contribute to the hardness; see Sub-figure 8(a–c). This means that if we take a look at a bar representing a class, the bar should be mostly colored in red and blue to indicate the positive and negative impact of feature groups on

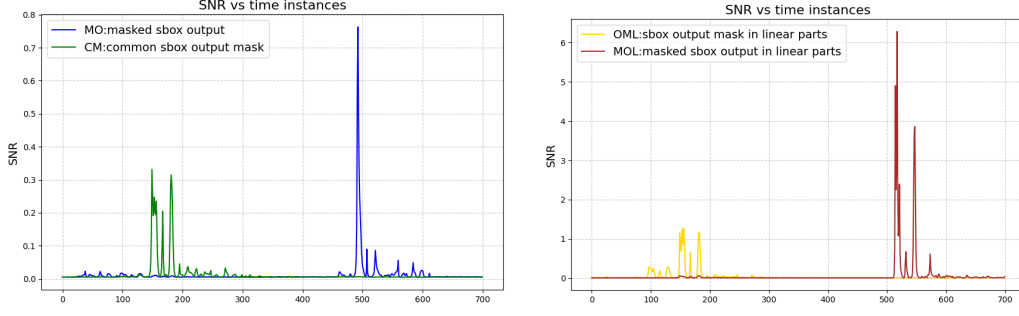


Figure 7: SNR vs. time samples in ASCAD dataset [BPS⁺17]. There are 700 time samples in ASCAD-fixed dataset [BPS⁺18] that based on the SNR, we can divide them into four groups: 1. MO (masked SBOX output, $Sbox(p[3] \oplus k[3]) \oplus r_{out}$) 2. CM (common SBOX output mask, r_{out}) 3. MOL (masked SBOX output in linear parts, $Sbox(p[3] \oplus k[3]) \oplus r[3]$) 4. OML (SBOX output mask in linear parts, $r[3]$).

hardness. Hence, per class, the fewer white sections in the groups, the more aligned the results are with the concepts in SCA. This is because white color indicates no significant influence of feature groups on the hardness (non-ideal). For instance, comparing the results obtained for QH and LO with MRE, MRE outperforms the others. This is because by looking at the first row of the Figure 7, we can see that the intensity of the contributions is higher while using the MRE as a metric. Now, looking at Sub-figure 8(d-f), MOL and OML are combined to form the group L-MSO (linear masked Sbox output). MO and CM are combined to create the group MSO (masked Sbox output). Again, MRE outperforms LO and QH; however, the white bars in the Sub-figure 8(f) highlight that combining the groups results in losing important information about the features and their impact on the hardness. This aligns with observations in SCA, where combining time samples can degrade performance by disrupting the temporal relationship between leakage traces and the implementation’s timing.

Moreover, in all sub-figures, the influence of different groups varies, although the variation is subtle (look at the black bars on the right-hand side of the sub-figures). Now, focusing on Sub-figure 8(c), an interesting effect can be observed that the MO and MOL groups are the most influential on the hardness, i.e., the useful information can be found in $Sbox(p[3] \oplus k[3] \oplus r[3])$ and $Sbox(p[3] \oplus k[3]) \oplus r_{out}$ (look at the black bars on the right-hand side of the plot). Although not surprising from the SCA perspective, this shows a strong tie between the notion of hardness and concepts in SCA. We can further investigate why MO’s impact on hardness is more pronounced. Compared to MOL, MO has a lower peak of SNR (see Figure 7 and the number of time samples in MO is higher than MOL, which explains why MO has a higher influence on the hardness. Seen from another perspective, the row corresponding to MOL includes more blue bars, meaning that MOL makes the

5.1 The Interplay between Hardness and SCA Metrics

In our next step, we investigate the relationship between hardness and SCA metrics. Particularly focusing on their relationship with rank prediction. In other words, we answer the question “does the hardness value help to explain the rank of the correct key for a trace?” To answer this, we apply a regressor chain that models the dependencies between the hardness and rank. To this end, the regressor chain sequentially predicts the hardness values and ranks. The hardness values were treated as the *first target (FT)* in the regressor chain model, predicted directly from the input traces. Once the hardness values were

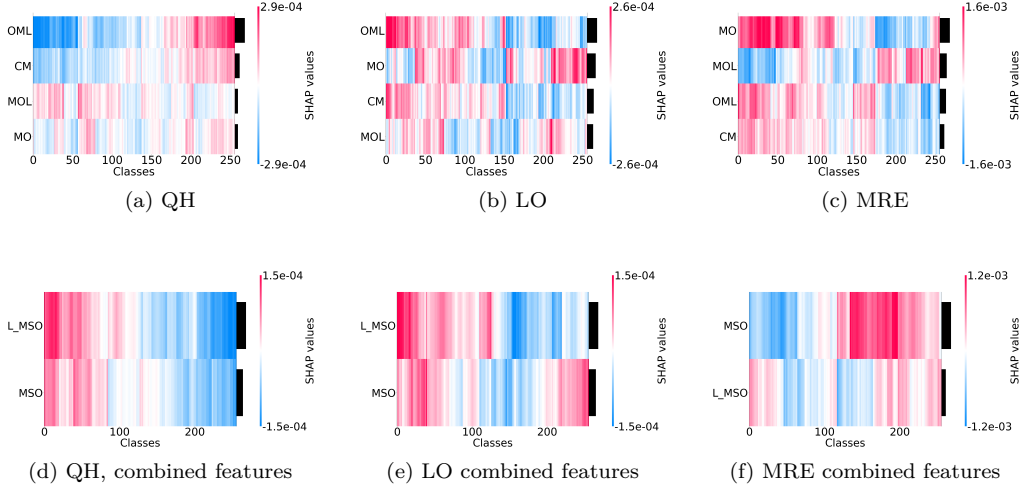


Figure 8: SHAP values to explore the influence of the time samples on the hardness values. There are 700 time instances in ASCAD-fixed dataset [BPS⁺17] that based on the SNR we can categorized them into four groups: 1. MO (masked SBOX output) 2. CM (common SBOX output mask) 3. MOL (masked SBOX output in linear parts) 4. OML (SBOX output mask in linear parts) cf. [BPS⁺18]. In these plots the y-axis is the SHAP values for time samples summed up in each group. For plots (d), (e), and (f), we combine the four groups into two: the linear masked Sbox output (L-MSO) and masked Sbox output (MSO), cf. Figure 7.

predicted, we used them as an additional feature to further predict the rank. This approach allowed us to explain the dependencies between the hardness and rank. By incorporating hardness as an intermediary prediction, we could examine its role as a meaningful metric that bridges input trace characteristics and the rank values.

In Figure 9, we present the results, highlighting the contribution of the hardness values (the first target-FT) to the rank. First and foremost, by taking a look at the summary plot (black bar plot on the right-hand side of sub-figures), it becomes clear that hardness strongly impacts the rank. The dependency between hardness and rank is even stronger than the value of time samples in different groups; see how FT is ordered at the top in all summary plots. Another interesting fact is that when relating the hardness and rank, QH and LO metrics fail to make a link between the values of time samples, hardness, and rank. In contrast to QH and LO, MRE could make a connection between them. Investigating Sub-figure 9(c) in more detail, the time samples in the OML and CM groups largely contribute to the rank as well as hardness (see the dark red part on the left-hand side of the plot). On the other hand, where some classes are easy, the rank is decided by the superposition of the impact of all time sample groups. For example, for class “150,” hardness is negatively related to the rank, while the rank positively depends on the time samples in OML and CM. On the other hand, considering class “100,” the rank is impacted by the time samples in MOL and MO.

6 Hardness from Class Distribution Perspective

As described in the previous sections, a high rank corresponding to poor SCA reflects that the correct class cannot be well separated from other classes. This has been reported in SCA-related literature, regardless of the model configuration and training. Even applying

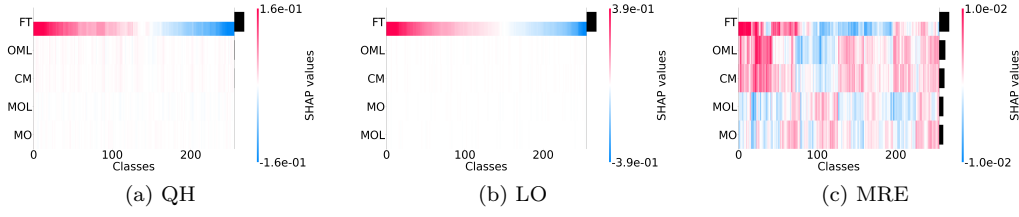


Figure 9: SHAP values for the regressor chain applied to the attack traces of the ASCAD dataset [BPS⁺18] with the time sample categories as in Figure 8. Here in the regressor chain, we also have an extra feature: the hardness value indicated as the first target (FT). In these plots, the x-axis shows the class number, whereas the y-axis represents the SHAP values.

sophisticated mechanisms [AGF23, PCP20, RWPP21, WPP22], this issue persists. This indicates that either the input data or the labeling context may cause classification hardness. The previous section has described that labeling cannot significantly impact the hardness; here, we delve into the nature of traces to explain why SCA is hard.

Generally speaking, if a trace is near a decision boundary, features extracted for that trace are closer to those of other classes, so even a well-chosen and well-trained model struggles to decide about the corresponding class [KDT19]. Furthermore, when classes cannot be well-separated in feature space due to being inherently similar, data points may overlap significantly, which prevents the NN from making a correct decision [BJY⁺17, Iva19]. This section investigates whether such cases happen in the context of SCA.

6.1 Visualization Techniques for Checking Input Ambiguity

Side-channel traces datasets contain samples represented in high-dimensional spaces, often with hundreds or thousands of features [BPS⁺18, NSGD12, PHJ⁺19b, CK09]. Analyzing or visualizing such data can be challenging due to the curse of dimensionality [VdMH08] and the difficulty of interpreting distances in high-dimensional spaces. In this work, the first visualization method we employ is t-distributed stochastic neighbor embedding (t-SNE) [VdMH08], a non-linear technique developed for the visualization of high-dimensional data in a two- or three-dimensional space. t-SNE models the similarity between data points as probability distributions and minimizes the divergence between these distributions in the original and embedded spaces.

A key motivation behind using a Student-t distribution is to address the crowding problem. In high-dimensional space, there is sufficient volume to accommodate moderately distant points. However, when visualizing the data points (traces) in two-dimensional space, the volume shrinks significantly, forcing those moderately distant points to be crowded together near the center. This compression distorts global and mid-range structures. The heavy tail of the Student-t distribution allows t-SNE to spread out distant points more effectively, mitigating the crowding issue and improving cluster separation. Due to its ability to preserve local structure while capturing meaningful global patterns, t-SNE has become a widely used technique for visualizing complex datasets in two or three dimensions.

Uniform manifold approximation and projection (UMAP) [MHM18] is the second technique that we apply. UMAP is a practical, scalable algorithm that relies on manifold learning applicable to real-world data and is competitive with t-SNE in visualization quality, while arguably preserving more of the *global structure* with superior runtime performance. UMAP uses local manifold approximations to construct a topological representation of the high-dimensional data. Given a low-dimensional representation of the data, a similar process is used to construct an equivalent topological representation. UMAP then

optimizes the layout of the data representation in the low-dimensional space to minimize the cross-entropy between the two topological representations.

6.2 Understanding Class Distributions in Feature Space

To understand whether the classes in SCA are inherently similar, we measure the difference between class distributions, not just individual samples. [GBR⁺12]. The task of distinguishing between distributions based on the distance between them is closely connected to classification [FGL⁺09]. Intuitively, if two distributions are easily distinguishable, they should also be easily classified. Hence, if side-channel traces cannot be classified easily, the distributions are indistinguishable, i.e., the distance between the class distributions should be low (non-ideal). This connection between class distinguishability and classification is particularly clear when using the maximum mean discrepancy (MMD) as the distance between distributions [DGL13, FGL⁺09]. MMD measures the distance between two distributions given a positive definite kernel¹ k , the MMD between two distributions p and q over a space X is defined as $\text{MMD}(p, q) = \|\mu - \nu\|_k$, where the kernel mean embeddings μ and ν are given by $\mu(\cdot) = \int k(x, \cdot) dP(x)$ and $\nu(\cdot) = \int k(x, \cdot) dQ(x)$. Here, μ and ν , often referred to as kernel mean maps, represent the embeddings of p and q into the reproducing RKHS associated with the kernel k . The norm $\|\cdot\|_k$ denotes the RKHS norm induced by this kernel. When applying MMD and RKHS distribution testing, the central question is which kernel to choose.

Radial kernels, especially radial basis function (RBF) kernels, have been considered a strong choice [HCL⁺03, GBR⁺12]. Unlike the linear kernels, the RBF kernel maps samples nonlinearly into a higher-dimensional feature space, enabling it to capture complex, non-linear relationships between input features and class labels. Therefore, besides being a general purpose, RBF kernels are suitable for SCA with non-linear separable traces and complex decision boundaries. More importantly, since closeness in traces implies semantic similarity, locality of RBFs is useful [See04, SFL11]. Here, locality means that the similarity between traces decays with increased distance. The most commonly applied RBF kernel is the Gaussian kernel, defined as

$$k(x_1, x_2) = \exp\left(-\frac{\|x_1 - x_2\|^2}{2\sigma^2}\right), \sigma > 0. \quad (15)$$

Defining $\gamma = 1/2\sigma^2$ as the *width* (or spread) yields $k(x_1, x_2) = \exp(-\gamma\|x_1 - x_2\|^2)$. γ should be tuned empirically. One of the most effective ways in this regard is to define $\gamma = 1/2 \cdot \text{median}(\|x_i - x_j\|^2)$, with the Euclidean distance $\|\cdot\|$ cf. [GBR⁺12].

Now, the question remains unanswered: how to set μ and ν . Fortunately, for a positive definite kernel k , MMD is a metric on the space of probability distributions, which can be approximated [SFL11, SKR22]. Given two sets of traces from two classes, $X_{c_1} = \{x_{1,1}, \dots, x_{n,1}\} \sim p$ and $X_{c_2} = \{x_{1,2}, \dots, x_{m,2}\} \sim q$, the test statistic is defined as:

$$\widehat{\text{MMD}}^2 = \frac{1}{n(n-1)m(m-1)} \sum_{i \neq i'} \sum_{j \neq j'} h(x_i, x_{i'}, y_j, y_{j'}), \quad (16)$$

where the kernel-based function h is given by $h(x_{i,1}, x_{j,1}, x_{i,2}, x_{j,2}) := k(x_{i,1}, x_{j,1}) - k(x_{i,1}, x_{j,2}) - k(x_{i,2}, x_{j,1}) + k(x_{i,2}, x_{j,2})$. $\widehat{\text{MMD}}^2$ measures the distinguishability between the distributions p and q based on pairwise kernel evaluations. It might be thought that computing $\widehat{\text{MMD}}^2$ is time-complex; however, it can be calculated efficiently in quadratic time [SKR22].

¹Kernel methods refer to embedding probability distributions into reproducing RKHS to handle higher-order statistics using linear techniques. MMD is the RKHS distance between two such mappings.

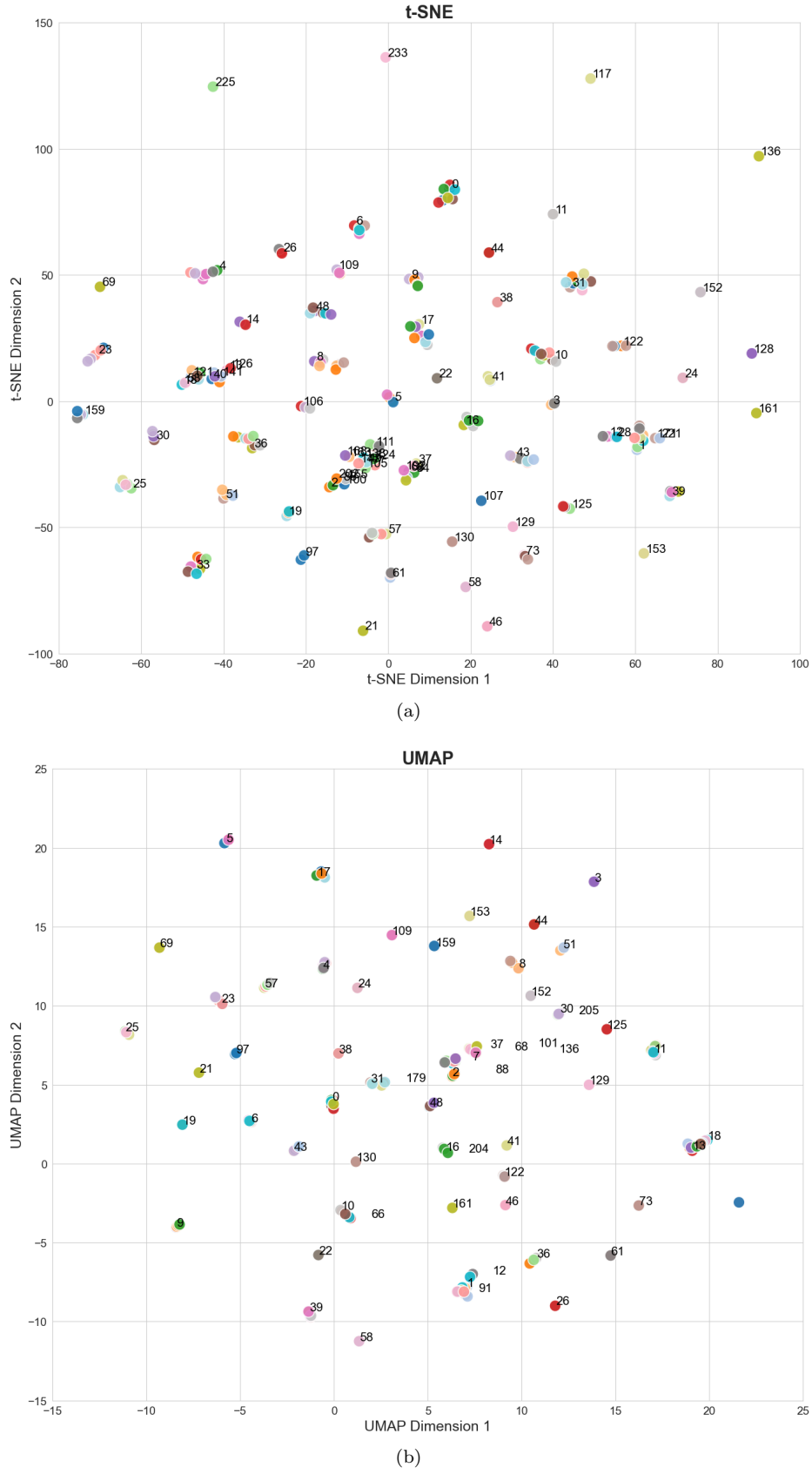


Figure 10: t-SNE [VdMH08] and UMAP [MHM18] illustration of close classes based on MMD. Here, each point corresponds to the vector of the MMD [GBR⁺12] of the traces of one class to other classes in ASCAD dataset[BPS⁺18]. The numbers on the points represent the label of the class, ranging from 0 to 255.

This estimate can be interpreted as the difference in inter-group similarities. When $\widehat{\text{MMD}}^2$ is large, the distributions p and q are significantly different, reflecting a higher complexity or distinguishability between the corresponding trace groups. In other words, a higher MMD suggests that the labels are easier to distinguish and classify.

How ambiguous are the attack traces? To assess the complexity of the side-channel attack (SCA) traces, we aim to measure the dissimilarity between different groups of labeled traces. First, we partition the attack traces by their label values and select exactly 23 traces from each group. This number corresponds to the maximum possible such that all classes have an equal number of traces in the ASCAD dataset [BPS⁺18]. This setup ensures a balanced representation across classes and enables a fair pairwise comparison.

To quantify the distance between the empirical distributions of traces from different classes, we employ MMD [GBR⁺12]. In doing so, we obtain a principled, kernel-based quantification of how separable the trace distributions are, thereby providing insight into the intrinsic hardness of the classification task. As there are 256 classes in the ASCAD dataset [BPS⁺18] after applying the MMD to the classes we have a matrix of size 256×256 whose the ij -th element is the distance between the class i and class j . In order to illustrate these distance values, we use t-SNE and UMAP as introduced in Section 6.1. Figure 10 shows the result. As can be seen in this figure, there is a significant overlap among data points across the 256 classes. In general, this behavior may not be uncommon when dealing with a high number of classes. However, it is also related to leakage traces. The point is that the underlying time sample does not provide sufficient class separability, i.e., traces share many similarities. This means that the traces are not easy to cluster, meaning the classes may not be well-separated in the feature space. Possibly, the reasons are high similarity between class distributions, noise, or insufficiently discriminative time samples. These observations highlight the importance of examining class structure in low-dimensional embeddings and the original feature space to understand the separability better. Therefore, next, we examine whether the hardness depends on MMD without reducing the dimension as needed for t-SNE and UMAP.

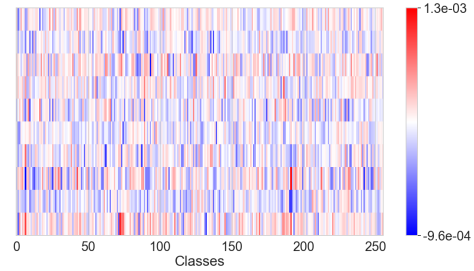


Figure 11: Dependency between MMD and hardness measured through MRE metric. The fewer white bars, the more dependent the two metrics are. On the x-axis, we have the class, whereas on the y-axis, SHAP values are depicted in a quantized fashion (50 levels of SHAP values) for the sake of visibility.

Relationship between MRE and MMD. We use the same approach as in Section 4 to explain the relationship between MMD and hardness. Concretely, we aim to answer whether the hardness depends on MMD. Based on our results in Section 5, MRE was our best hardness metric; hence, we used this metric as the target feature for the results in this section. Figure 11 shows the SHAP values indicating the dependency between MMD and MRE. Interestingly enough, most bars are not white, indicating either MMD ease hardness (blue bars), or contribute to the hardness positively. As can be observed in the plot, the hardness of each class depends on its closeness to other classes. If a class is in proximity to other classes, the SHAP values for those classes are colored red. If two classes are disjoint, but their trace time samples have similarities, MMD helps to resolve the hardness. There are also cases where MMD does not have an influence, i.e., the SHAP value is 0. Under such scenarios, neither class distribution overlaps, nor do the traces share a great deal of similarity. In our analysis, less than 1/3 of classes exhibit such behavior. Our analysis shows that 43618 out of 65536 SHAP values were close to zero and had minimal contribution.

7 A Short Note on the Importance of Hardness

As mentioned before, SCA-related literature has noticed the importance of defining a “hard” to classify sample. Nevertheless, we have not found a thoroughly systematic analysis devoted to this. The recent paper in TCHES Vol. 2025, No. 2 has investigated the application of statistical analysis in defining a new performance evaluation metric, so-called leading degree [ZL25]. Their methodology makes multiple assumptions to define the metric. The distribution of scores is approximated by (linearly transformed) $\chi^2(1)$, i.e., Chi-square distribution with 1 degree of freedom. Based on this, it has been proved that there is a negative correlation between the coefficient of variation² of the probability of distinguishing the correct key from wrong ones and its average over q attack traces. The authors have also stated that in practice, this may not hold (See Section 4.2 in [ZL25]); hence, they have suggested using the reciprocal of the coefficient of variation, which can have a positive correlation with the distinguishability probability.

Remark 1. They also mentioned that residing on a chi-square distribution would tackle the requirement for a large number of traces to approximate the distributions. In fact, the distribution $\chi^2(1)$ is the distribution of the square of a standard normal variable, i.e, it is constructed from only one standard normal distribution. The coefficient of variation exhibits instability when applied to highly skewed distributions such as $\chi^2(1)$, particularly in the context of small sample sizes. Therefore, the argument about requiring only a small number of traces would not be valid. Therefore, the feasibility of evaluating attacks’ “difficulty” through the reciprocal of the coefficient of variation of $y(k^*) - y(k)$ needs to be revisited.

Remark 2. If the output of a softmax function follows (a linearly transformed) $\chi^2(1)$, the logits should follow a $\log - \chi^2$ distribution $z_i \sim \log(\chi^2(1))$. $\log - \chi^2$ distribution is highly skewed, heavy-tailed, and similar in shape to a log-normal. Suppose that the argument about the output of a softmax function is true. This implies that the model is extremely overconfident: producing large logits for a few classes, while assigning very low probability to most others (very negative logits) [GPSW17]. Moreover, as the logit distribution is non-Gaussian and heavy-tailed, the network’s internal representations or weight configurations may occasionally generate disproportionately large activations. Such behavior may stem from factors including imbalanced or extreme input features, suboptimal weight initialization, saturation effects in activation functions, or excessive model capacity due to overparameterization— in short, imperfect model selection [GB10, HZRS15]. Distribution shift or overfitting could be another relevant cause. In summary, if the output of the softmax function follows a $\chi^2(1)$ distribution, this may indicate deficiencies in the profiling setup. Specifically, evaluating attack performance using a $\chi^2(1)$ distribution may be ill-posed or inappropriate.

8 Conclusion

In this paper, we investigated the concept of hardness in the classification task in SCA. We adapted multiple hardness metrics from the ML domain, namely query hardness (QH), log-odds (LO), and matrix-based Rényi entropy (MRE), to quantitatively characterize attack difficulty in the SCA context (to our knowledge, this is the first application of these metrics in SCA). Using these metrics, we obtained a fine grained view of how “hard” each trace or class is, enabling deeper insight into the performance of the key recovery. In addition, we incorporated XAI techniques by leveraging SHAP values to interpret the relationship between the label bits and the hardness values. The SHAP-based analysis provided a framework for our hardness measures, confirming that label bits or encoding

²The coefficient of variation of a random variable δ is $CV[\delta] = \sqrt{\text{Var}(\delta)/\mathbb{E}[\delta]}$.

contribute negligibly to the observed hardness. This result held consistently across both profiling and attack phases in our study, suggesting that factors other than label encoding are responsible for variations in classification difficulty.

This led us to take a look at other aspects of SCA, namely time samples in the traces to explain the influence on classification hardness. Here, we observed that high-SNR regions of the traces contribute positively to the hardness values. Our results also show the advantage of the MRE over other metrics while analyzing the contribution of the time instances.

References

- [ABB⁺20] Melissa Azouaoui, Davide Bellizia, Ileana Buhan, Nicolas Debande, Sébastien Duval, Christophe Giraud, Éliane Jaulmes, François Koeune, Elisabeth Oswald, François-Xavier Standaert, et al. A systematic appraisal of side channel evaluation strategies. In *International Conference on Research in Security Standardisation*, pages 46–66. Springer, 2020.
- [AGF23] Rabin Y Acharya, Fatemeh Ganji, and Domenic Forte. Information theory-based evolution of neural networks for side-channel analysis. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, pages 401–437, 2023.
- [AIS21] AISyLab. Supplementary code: The best of two worlds: Deep learning-assisted template attack. [Online]<https://github.com/AISyLab/Triplet-attack> [Accessed: May.19, 2025], 2021.
- [BCGR24] Julien Béguinot, Wei Cheng, Sylvain Guilley, and Olivier Rioul. Be my guesses: The interplay between side-channel leakage metrics. *Microprocessors and Microsystems*, 107:105045, 2024.
- [BFT17] Peter L Bartlett, Dylan J Foster, and Matus J Telgarsky. Spectrally-normalized margin bounds for neural networks. *Advances in neural information processing systems*, 30, 2017.
- [BGH⁺17] Nicolas Bruneau, Sylvain Guilley, Annelie Heuser, Damien Marion, and Olivier Rioul. Optimal side-channel attacks for multivariate leakages and multiple models. *Journal of Cryptographic Engineering*, 7:331–341, 2017.
- [Bha06] Rajendra Bhatia. Infinitely divisible matrices. *The American Mathematical Monthly*, 113(3):221–235, 2006.
- [BJY⁺17] Alsallakh Bilal, Amin Jourabloo, Mao Ye, Xiaoming Liu, and Liu Ren. Do convolutional neural networks learn class hierarchy? *IEEE transactions on visualization and computer graphics*, 24(1):152–162, 2017.
- [BN06] Christopher M Bishop and Nasser M Nasrabadi. *Pattern recognition and machine learning*, volume 4. Springer, 2006.
- [BPS⁺17] Ryad Benadjila, Emmanuel Prouff, Rémi Strullu, Eleonora Cagli, and Cécile Dumas. ASCAD: the ATMega8515 SCA traces databases (fixed key). [Online]https://github.com/ANSSI-FR/ASCAD/tree/master/ATMEGA_AES_v1/ATM_AES_v1_fixed_key [Accessed: May.19, 2025], 2017.
- [BPS⁺18] Ryad Benadjila, Emmanuel Prouff, Rémi Strullu, Eleonora Cagli, and Cécile Dumas. Study of deep learning techniques for side-channel analysis and introduction to ascad database. 2018.
- [BW95] Brad J Bushman and Morgan C Wang. A procedure for combining sample correlation coefficients and vote counts to obtain an estimate and a confidence interval for the population correlation coefficient. *Psychological Bulletin*, 117(3):530, 1995.
- [CDB98] David M Corey, William P Dunlap, and Michael J Burke. Averaging correlations: Expected values and bias in combined pearson rs and fisher’s z transformations. *The Journal of general psychology*, 125(3):245–261, 1998.

- [Cha18] John M Chambers. *Graphical methods for data analysis*. Chapman and Hall/CRC, 2018.
- [CK09] Jean-Sébastien Coron and Ilya Kizhvatov. An efficient method for random delay generation in embedded software. In *International Workshop on Cryptographic Hardware and Embedded Systems*, pages 156–170. Springer, 2009.
- [Cov99] Thomas M Cover. *Elements of information theory*. John Wiley & Sons, 1999.
- [CRR02] Suresh Chari, Josyula R Rao, and Pankaj Rohatgi. Template attacks. In *Intrl. Workshop on Cryptographic Hardware and Embedded Systems*, pages 13–28. Springer, 2002.
- [DCGRP19] Eloi De Cherisey, Sylvain Guilley, Olivier Rioul, and Pablo Piantanida. An information-theoretic model for side-channel attacks in embedded hardware. In *2019 IEEE International Symposium on Information Theory (ISIT)*, pages 310–315. IEEE, 2019.
- [DCRS19] Guneet S Dhillon, Pratik Chaudhari, Avinash Ravichandran, and Stefano Soatto. A baseline for few-shot image classification. *arXiv preprint arXiv:1909.02729*, 2019.
- [DGL13] Luc Devroye, László Györfi, and Gábor Lugosi. *A probabilistic theory of pattern recognition*, volume 31. Springer Science & Business Media, 2013.
- [DPRS11] Julien Doget, Emmanuel Prouff, Matthieu Rivain, and François-Xavier Standaert. Univariate side channel attacks and leakage modeling. *Journal of Cryptographic Engineering*, 1(2):123, 2011.
- [FGL⁺09] Kenji Fukumizu, Arthur Gretton, Gert Lanckriet, Bernhard Schölkopf, and Bharath K Sriperumbudur. Kernel choice and classifiability for rkhs embeddings of probability distributions. *Advances in neural information processing systems*, 22, 2009.
- [GB10] Xavier Glorot and Yoshua Bengio. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics*, pages 249–256. JMLR Workshop and Conference Proceedings, 2010.
- [GBC16] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT press, 2016.
- [GBR⁺12] Arthur Gretton, Karsten M Borgwardt, Malte J Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *The Journal of Machine Learning Research*, 13(1):723–773, 2012.
- [Gér22] Aurélien Géron. *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems*. " O'Reilly Media, Inc.", 2022.
- [GLRP06] Benedikt Gierlichs, Kerstin Lemke-Rust, and Christof Paar. Templates vs. stochastic methods: A performance analysis for side channel cryptanalysis. In *International Workshop on Cryptographic Hardware and Embedded Systems*, pages 15–29. Springer, 2006.
- [GMO01] Karine Gandolfi, Christophe Mourtel, and Francis Olivier. Electromagnetic analysis: Concrete results. In *Cryptographic Hardware and Embedded Systems—CHES 2001*, pages 251–261. Springer, 2001.
- [GP13] Luis G Sanchez Giraldo and Jose C Principe. Rate-distortion auto-encoders. *arXiv preprint arXiv:1312.7381*, 2013.
- [GPSW17] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR, 2017.
- [GRP14] Luis Gonzalo Sanchez Giraldo, Murali Rao, and Jose C Principe. Measures of entropy from data using infinitely divisible kernels. *Trans. on Information Theory*, 61(1):535–548, 2014.

- [GW68] Ramanathan Gnanadesikan and Martin B Wilk. Probability plotting methods for the analysis of data. *Biometrika*, 55(1):1–17, 1968.
- [HCL⁺03] Chih-Wei Hsu, Chih-Chung Chang, Chih-Jen Lin, et al. A practical guide to support vector classification, 2003.
- [HK70] Arthur E Hoerl and Robert W Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.
- [HSW89] Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- [HSW01] Peter E Hart, David G Stork, and John Wiley. Pattern classification. 2001.
- [HTFF09] Trevor Hastie, Robert Tibshirani, Jerome H Friedman, and Jerome H Friedman. *The elements of statistical learning: data mining, inference, and prediction*, volume 2. Springer, 2009.
- [HZRS15] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the IEEE international conference on computer vision*, pages 1026–1034, 2015.
- [Iva19] Cristian Ivan. Convolutional neural networks on randomized data. In *CVPR Workshops*, pages 1–8, 2019.
- [Jai10] Anil K Jain. Data clustering: 50 years beyond k-means. *Pattern recognition letters*, 31(8):651–666, 2010.
- [JKG18] Heinrich Jiang, Been Kim, Melody Guan, and Maya Gupta. To trust or not to trust a classifier. *Advances in neural information processing systems*, 31, 2018.
- [JWH⁺13] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, et al. *An introduction to statistical learning*, volume 112. Springer, 2013.
- [KB19] Mahmut Kaya and Hasan Şakir Bilge. Deep metric learning: A survey. *Symmetry*, 11(9):1066, 2019.
- [KDT19] Hamid Karimi, Tyler Derr, and Jiliang Tang. Characterizing the decision boundary of deep neural networks. *arXiv preprint arXiv:1912.11460*, 2019.
- [KJJ99] Paul Kocher, Joshua Jaffe, and Benjamin Jun. Differential power analysis. In *Advances in Cryptology—CRYPTO’99: 19th Annual International Cryptology Conference Santa Barbara, California, USA, August 15–19, 1999 Proceedings 19*, pages 388–397. Springer, 1999.
- [LBM14] Liran Lerman, Gianluca Bontempi, and Olivier Markowitch. Power analysis attack: an approach based on machine learning. *International Journal of Applied Cryptography*, 3(2):97–115, 2014.
- [LM18] Liran Lerman and Olivier Markowitch. Efficient profiled attacks on masking schemes. *IEEE Transactions on Information Forensics and Security*, 14(6):1445–1454, 2018.
- [LPMS18] Liran Lerman, Romain Poussier, Olivier Markowitch, and François-Xavier Standaert. Template attacks versus machine learning revisited and the curse of dimensionality in side-channel analysis: extended version. *Journal of Cryptographic Engineering*, 8(4):301–313, 2018.
- [LTR17] Henry W Lin, Max Tegmark, and David Rolnick. Why does deep and cheap learning work so well? *Journal of Statistical Physics*, 168:1223–1247, 2017.
- [LW22] Stjepan Picek Lichao Wu, Guilherme Perin. AutoSCA: automated hyperparameter tuning for deep learning-based side-channel analysis. [Online]<https://github.com/AISyLab/AutoSCA> [Accessed: May.19, 2025], 2022.
- [Mac67] James MacQueen. Some methods for classification and analysis of multivariate observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Statistics*, volume 5, pages 281–298. University of California press, 1967.

- [MCLS23] Loïc Masure, Valence Cristiani, Maxime Lecomte, and François-Xavier Standaert. Don't learn what you already know: Scheme-aware modeling for profiling side-channel analysis against masking. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, pages 32–59, 2023.
- [MDM16] Zdenek Martinasek, Petr Dzurenda, and Lukas Malina. Profiling power analysis attack based on mlp in dpa contest v4. 2. In *2016 39th International Conference on Telecommunications and Signal Processing (TSP)*, pages 223–226. IEEE, 2016.
- [MDP20] Loïc Masure, Cécile Dumas, and Emmanuel Prouff. A comprehensive study of deep learning for side-channel analysis. *IACR Trans. on Cryptographic Hardware and Embedded Systems*, pages 348–375, 2020.
- [MHM18] Leland McInnes, John Healy, and James Melville. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.
- [MKH19] Rafael Müller, Simon Kornblith, and Geoffrey E Hinton. When does label smoothing help? *Advances in neural information processing systems*, 32, 2019.
- [MMOS16] Daniel P Martin, Luke Mather, Elisabeth Oswald, and Martijn Stam. Characterisation and estimation of the key rank distribution in the context of side channel evaluations. In *Advances in Cryptology—ASIACRYPT 2016: 22nd International Conference on the Theory and Application of Cryptology and Information Security, Hanoi, Vietnam, December 4–8, 2016, Proceedings, Part I 22*, pages 548–572. Springer, 2016.
- [MO23] Thomas Marquet and Elisabeth Oswald. A comparison of multi-task learning and single-task learning approaches. In *International Conference on Applied Cryptography and Network Security*, pages 121–138. Springer, 2023.
- [MPP16] Housseem Maghrebi, Thibault Portigliatti, and Emmanuel Prouff. Breaking cryptographic implementations using deep learning techniques. In *Intrnl. Conf. on Security, Privacy, and Applied Cryptography Engineering*, pages 3–26. Springer, 2016.
- [MWX⁺22] Sina Mohseni, Haotao Wang, Chaowei Xiao, Zhiding Yu, Zhangyang Wang, and Jay Yadawa. Taxonomy of machine learning safety: A survey and primer. *ACM Computing Surveys*, 55(8):1–38, 2022.
- [MZ13] Zdenek Martinasek and Vaclav Zeman. Innovative method of the power analysis. *Radioengineering*, 22(2):586–594, 2013.
- [NG24] Seyedmohammad Nouraniboosjin and Fatemeh Ganji. Too hot to be true: Temperature calibration for higher confidence in nn-assisted side-channel analysis. *Cryptology ePrint Archive*, 2024.
- [NG25] Seyedmohammad Nouraniboosjin and Fatemeh Ganji. Uncertainty estimation in neural network-enabled side-channel analysis and links to explainability. *Cryptology ePrint Archive*, 2025.
- [NSGD12] Maxime Nassar, Youssef Souissi, Sylvain Guilley, and Jean-Luc Danger. Rsm: A small and fast countermeasure for aes, secure against 1st and 2nd-order zero-offset scas. In *2012 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pages 1173–1178. IEEE, 2012.
- [PCP20] Guilherme Perin, Łukasz Chmielewski, and Stjepan Picek. Strength in numbers: Improving generalization with ensembles in machine learning-based profiled side-channel analysis. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, pages 337–364, 2020.
- [PHJ⁺19a] Stjepan Picek, Annelie Heuser, Alan Jovic, Shivam Bhasin, and Francesco Regazzoni. The curse of class imbalance and conflicting metrics with machine learning for side-channel evaluations. *IACR Trans. on Cryptographic Hardware and Embedded Systems*, 2019(1):1–29, 2019.
- [PHJ⁺19b] Stjepan Picek, Annelie Heuser, Alan Jovic, Shivam Bhasin, and Francesco Regazzoni. The curse of class imbalance and conflicting metrics with machine learning for side-channel evaluations. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, pages 209–237, 2019.

- [PPM⁺21] Stjepan Picek, Guilherme Perin, Luca Mariot, Lichao Wu, and Lejla Batina. SoK: Deep learning-based physical side-channel analysis. *IACR Cryptol. ePrint Arch.*, 2021/1092, 2021.
- [PR09] Emmanuel Prouff and Matthieu Rivain. Theoretical and practical aspects of mutual information based side channel analysis. In *Applied Cryptography and Network Security: 7th International Conference, ACNS 2009, Paris-Rocquencourt, France, June 2-5, 2009. Proceedings* 7, pages 499–518. Springer, 2009.
- [PR10] Emmanuel Prouff and Matthieu Rivain. Theoretical and practical aspects of mutual information-based side channel analysis. *International Journal of Applied Cryptography*, 2(2):121–138, 2010.
- [PWP22] Guilherme Perin, Lichao Wu, and Stjepan Picek. Exploring feature selection scenarios for deep learning-based side-channel analysis. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, pages 828–861, 2022.
- [PYW⁺17] Sihang Pu, Yu Yu, Weijia Wang, Zheng Guo, Junrong Liu, Dawu Gu, Lingyun Wang, and Jie Gan. Trace augmentation: What can be done even before preprocessing in a profiled sca? In *International Conference on Smart Card Research and Advanced Applications*, pages 232–247. Springer, 2017.
- [RBAF24] Unai Rioja, Lejla Batina, Igor Armendariz, and Jose Luis Flores. Keep it unbiased: a comparison between estimation of distribution algorithms and deep learning for human interaction-free side-channel analysis. *Journal of Cryptographic Engineering*, 14(3):499–511, 2024.
- [RBFA21] Unai Rioja, Lejla Batina, Jose Luis Flores, and Igor Armendariz. Auto-tune pois: Estimation of distribution algorithms for efficient side-channel analysis. *Computer Networks*, 198:108405, 2021.
- [RHMM25] Nathan Rousselot, Karine Heydemann, Loïc Masure, and Vincent Migairou. Scoop: An optimizer for profiling attacks against higher-order masking. *Cryptology ePrint Archive*, 2025.
- [Riv08] Matthieu Rivain. On the exact success rate of side channel analysis in the gaussian model. In *International Workshop on Selected Areas in Cryptography*, pages 165–183. Springer, 2008.
- [Rou87] Peter J Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of computational and applied mathematics*, 20:53–65, 1987.
- [RWPP21] Jorai Rijdsdijk, Lichao Wu, Guilherme Perin, and Stjepan Picek. Reinforcement learning for hyperparameter tuning in deep learning-based side-channel analysis. *Cryptol. ePrint Arch.*, Report 2021/071, 2021.
- [Seb09] George AF Seber. *Multivariate observations*. John Wiley & Sons, 2009.
- [See04] Matthias Seeger. Gaussian processes for machine learning. *International journal of neural systems*, 14(02):69–106, 2004.
- [SFL11] Bharath K Sriperumbudur, Kenji Fukumizu, and Gert RG Lanckriet. Universality, characteristic kernels and rkhs embedding of measures. *Journal of Machine Learning Research*, 12(7), 2011.
- [SGG16] Abhinav Shrivastava, Abhinav Gupta, and Ross Girshick. Training region-based object detectors with online hard example mining. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 761–769, 2016.
- [SKR22] Shubhanshu Shekhar, Ilmun Kim, and Aaditya Ramdas. A permutation-free kernel two-sample test. *Advances in Neural Information Processing Systems*, 35:18168–18180, 2022.
- [SMY09] François-Xavier Standaert, Tal G Malkin, and Moti Yung. A unified framework for the analysis of side-channel key recovery attacks. In *Annual Intrnl. Conf. on the Theory and Applications of Cryptographic Techniques*, pages 443–461. Springer, 2009.

- [SREG⁺25] Ahmed M Salih, Zahra Raisi-Estabragh, Ilaria Boscolo Galazzo, Petia Radeva, Steffen E Petersen, Karim Lekadir, and Gloria Menegaz. A perspective on explainable artificial intelligence methods: Shap and lime. *Advanced Intelligent Systems*, 7(1):2400304, 2025.
- [SSSS17] Shai Shalev-Shwartz, Ohad Shamir, and Shaked Shammah. Failures of gradient-based deep learning. In *International Conference on Machine Learning*, pages 3067–3075. PMLR, 2017.
- [SVCO⁺10] Francois-Xavier Standaert, Nicolas Veyrat-Charvillon, Elisabeth Oswald, Benedikt Gierlichs, Marcel Medwed, Markus Kasper, and Stefan Mangard. The world is not enough: Another look on second-order dpa. In *Advances in Cryptology-ASIACRYPT 2010: 16th International Conference on the Theory and Application of Cryptology and Information Security, Singapore, December 5-9, 2010. Proceedings 16*, pages 112–129. Springer, 2010.
- [Tho53] Robert L Thorndike. Who belongs in the family? *Psychometrika*, 18(4):267–276, 1953.
- [TSK16] Pang-Ning Tan, Michael Steinbach, and Vipin Kumar. *Introduction to data mining*. Pearson Education India, 2016.
- [VCMKS12] Nicolas Veyrat-Charvillon, Marcel Medwed, Stéphanie Kerckhof, and François-Xavier Standaert. Shuffling against side-channel attacks: A comprehensive study with cautionary note. In *Advances in Cryptology-ASIACRYPT 2012: 18th International Conference on the Theory and Application of Cryptology and Information Security, Beijing, China, December 2-6, 2012. Proceedings 18*, pages 740–757. Springer, 2012.
- [VdMH08] Laurens Van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of machine learning research*, 9(11), 2008.
- [WAGP20] Lennert Wouters, Victor Arribas, Benedikt Gierlichs, and Bart Preneel. Revisiting a methodology for efficient cnn architectures in profiling attacks. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, pages 147–168, 2020.
- [WPP21] Lichao Wu, Guilherme Perin, and Stjepan Picek. The best of two worlds: Deep learning-assisted template attack. *IACR Cryptol. ePrint Arch.*, 2021/959, 2021.
- [WPP22] Lichao Wu, Guilherme Perin, and Stjepan Picek. I choose you: Automated hyperparameter tuning for deep learning-based side-channel analysis. *IEEE Transactions on Emerging Topics in Computing*, 2022.
- [WWK⁺23] Lichao Wu, Léo Weissbart, Marina Krček, Huimin Li, Guilherme Perin, Lejla Batina, and Stjepan Picek. Label correlation in deep learning-based side-channel analysis. *Ieee transactions on information forensics and security*, 18:3849–3861, 2023.
- [YP19] Shujian Yu and José C Príncipe. Simple stopping criteria for information theoretic feature selection. *Entropy*, 21(1):99, 2019.
- [YPB24] Trevor Yap, Stjepan Picek, and Shivam Bhasin. Occpois: Points of interest based on neural network’s key recovery in side-channel analysis through occlusion. In *International Conference on Cryptology in India*, pages 3–28. Springer, 2024.
- [YWJP20] Shujian Yu, Kristoffer Wickstrøm, Robert Jenssen, and José C Príncipe. Understanding convolutional neural networks with information theory: An initial exploration. *Trans. on Neural Networks and Learning Systems*, 2020.
- [YZSH20] Zhongqi Yue, Hanwang Zhang, Qianru Sun, and Xian-Sheng Hua. Interventional few-shot learning. *Advances in neural information processing systems*, 33:2734–2746, 2020.
- [ZBHV20] Gabriel Zaid, Lilian Bossuet, Amaury Habrard, and Alexandre Venelli. Methodology for efficient cnn architectures in profiling attacks. *IACR Trans. on Cryptographic Hardware and Embedded Systems*, 2020(1):1–36, 2020.
- [ZL25] Junfan Zhu and Jiqiang Lu. Leading degree: A metric for model performance evaluation and hyperparameter tuning in deep learning-based side-channel analysis. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2025(2):333–361, 2025.